

Article

Pesticide Use, Regulation, and Policies in Indian Agriculture

A. Amarender Reddy ^{1,*}, Meghana Reddy ² and Vartika Mathur ^{3,*}

¹ ICAR-National Institute of Biotic Stress Management (ICAR-NIBSM), Raipur 493225, India

² ICAR-Agricultural Technology Applications Institute (ICAR-ATARI), Bangalore 560024, India; meghareddy156@gmail.com

³ Department of Zoology, Sri Venkateswara College, University of Delhi, New Delhi 110021, India

* Correspondence: amarender.reddy@icar.gov.in (A.A.R.); vmathur@svc.ac.in (V.M.)

† These authors contributed equally to this work.

Abstract: This research paper presents a comprehensive analysis of pesticide use in global and Indian agriculture, focusing on the mounting food security challenges due to population growth and the increased demand for food and fiber crops. While pesticides are crucial in mitigating losses due to pests, diseases, and weeds, increasing apprehension regarding their adverse effects on human and environment health necessitates a critical examination of their usage patterns. Despite India's relatively low per-hectare pesticide usage of 0.4 kg compared to China's 1.83 kg, issues with pesticide residue contamination in the food chain require urgent attention. Additionally, significant regional disparities in pesticide application highlight the need for more uniform and sustainable agricultural practices. Therefore, this study examines the evolving trends in global and Indian pesticide application, providing a comprehensive analysis of the shifting dynamics across various Indian states and crop varieties. Furthermore, it analyzes various pesticide categories and their respective market shares, providing insights into production and export patterns. Our research also explores regulatory frameworks aimed at optimizing pesticide use while minimizing detrimental effects on human health and the environment.

Keywords: pesticides; India; pesticide residues; regulation; policies



Citation: Reddy, A.A.; Reddy, M.; Mathur, V. Pesticide Use, Regulation, and Policies in Indian Agriculture. *Sustainability* **2024**, *16*, 7839. <https://doi.org/10.3390/su16177839>

Academic Editors: Renato Almeida Sarmiento and Ricardo Siqueira Da Silva

Received: 26 May 2024

Revised: 5 August 2024

Accepted: 5 September 2024

Published: 9 September 2024



Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Agriculture is the backbone of the Indian economy, contributing 18 per cent of the total gross domestic product (GDP) and providing employment for about 45% of the population. Assuring food security for more than 1.27 bn Indian residents with decreasing cultivable land resources is a difficult task [1]. Global agriculture incurs approximately 20–30% annual losses due to pests, diseases, and weeds [2–9]. Pesticides are used to control pests, weeds, and diseases in agriculture (see Appendix A, Tables A1–A3 for detailed crop losses due to insects, diseases, and weeds). A number of chemical and biological pesticides are used to minimize crop losses. Up to the 1950s, pesticide usage was minimal worldwide. However, the beginning of green revolution in the 1960s marked a dramatic increase in pesticide application globally, including in India, where their usage surged over a hundredfold, resulting in a significant impact on the environment and human health [10]. In 2020, approximately 3.39 million tons of pesticides were used in agriculture globally, with India accounting for 61,702 tons [11].

There is a growing body of literature on the excessive use of pesticides in agriculture, both in India and globally [8,12–15]. However, despite the widespread use of pesticides in India, there is a lack of comprehensive analysis regarding the extent of different pesticide use habits in India, their geographical patterns, and crop-specific intensity. Hence, there is an urgent need to address this gap. This paper aims to fill this important gap in the literature by conducting an in-depth examination of pesticide usage patterns and consumption across different crops and regions. By making use of the secondary data from input surveys and

other sources collected by Government of India over the past 40 years, this study provides a detailed analysis of these trends.

2. Objectives and Methodology

The primary objective of this study is to investigate the implications of pesticide usage patterns for Indian agriculture, focusing on both the benefits and risks associated with pesticide use. The study thus aims to elucidate the evolution of pesticide usage by examining the trends in pesticide usage over the years, across different regions and crop types, and in both irrigated and unirrigated areas in India. It also aims to analyze the regulatory landscape by evaluating the policy issues related to the Pesticides Act, including provisions for the proper and optimal use of pesticides, the banning of harmful pesticides, and the effectiveness of these regulations. In addition, it assesses the impact on food safety and the environment by exploring the incidence of pesticide residues in the food chain and their potential effects on human health and the environment. Finally, the study provides recommendations for sustainable practices to enhance the sustainability of pesticide use in Indian agriculture.

To achieve these aims, our study utilized secondary data from various official sources, including the Food and Agriculture Organization Statistics (FAOSTAT) and Indian government publications such as “Agriculture at a Glance” and the “All India Report on Input Survey” for the years 2011–2012 (Fiscal year 2011–2012 in India, which begins on 1 April 2011, and ends on 31 March 2012) and 2015–2016. Additional sources included reports from the States/UTs Zonal Conferences on Inputs of Plant Protection, the “Compendium of Environment Statistics India, 2011”, and websites like those of the Food Safety and Standards Authority of India (FSSAI) and the Directorate of Plant Protection, Quarantine & Storage (DPPQS), Government of India [11,16–18]. The latest data available up to 2016–2017 was incorporated. For trend analysis, a temporal study of pesticide usage data was performed to identify trends and patterns across different regions, crops, and types of agricultural practices (irrigated vs. unirrigated). Moreover, regional disparities in pesticide use were determined, identifying areas with high and low usage and correlating these patterns with crop types and pest pressures.

We also performed an impact analysis of pesticide usage by assessing the prevalence of pesticide residues in the food supply and their potential health impacts, using data from the FSSAI and other relevant bodies. A critical review of the Pesticides Act and related regulatory frameworks was performed by assessing the adequacy and effectiveness of existing policies in regulating pesticide use and ensuring safety. The environmental consequences of pesticide use, including the potential contamination of soil and water resources, were also analyzed in this study. Based on the findings, the study proposes recommendations for improving pesticide management practices, enhancing regulatory frameworks, and promoting safer and more sustainable agricultural practices.

3. Results

3.1. Pesticide Use in the World and in India

Figure 1 compares the data on pesticide utilization globally and in India in the year 2020. Worldwide, the total consumption amounted to 3.39 million tons, while in India, it was 61,702 tons (FAOSTAT, Rome, Italy, 2024). There are notable differences in the types of pesticides used. Globally, herbicides constitute the largest share at approximately 50%, followed by fungicides and bactericides at 22.5% and insecticides at 20.4%. Plant growth regulators represent a minimal share of just 1.2%. Conversely, in India, insecticides dominate, with a share of 51.4%, followed by fungicides and bactericides at 32.6% and herbicides at 15.8%. The lower usage of herbicides in India can be attributed to availability of cheap labor and the limited adoption of herbicide-tolerant GMO crops. In contrast, many developed countries permit the use of herbicide-tolerant varieties and incur higher labor costs for weed management. The consumption of mineral oils as fungicides and insecticides for seed treatments, rodenticides, and disinfectants is limited. Although seed

treatment is highly effective for many crops, there is a need to increase awareness about its benefits in India.

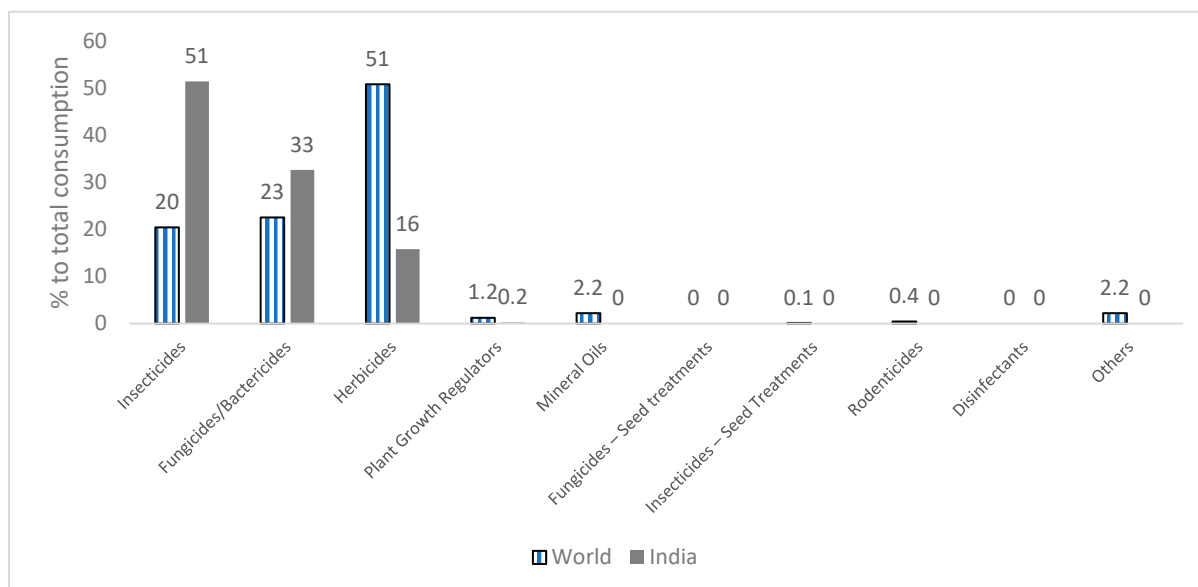


Figure 1. Various types of pesticide consumption in the world and in India, 2020. Note: calculated from FAOSTAT data available at <https://www.fao.org/faostat/en/#data/RP>, accessed on 16 June 2024 [19].

Figure 2 depicts the trends in pesticide usage per unit area (kg/ha) in India and comparable countries such as Brazil, China, Germany, and the USA. Across the examined period, India consistently exhibited lower pesticide usage compared to the other nations. Particularly in Brazil, pesticide usage surged from 1.1 kg/ha in 1990 to 10.9 kg/ha in 2021. Similarly, Germany's pesticide usage increased from 2.5 to 4.1 kg/ha during the same timeframe. In the USA, it rose from 2.14 to 2.85 kg/ha, and in China, it increased from 1.1 kg/ha to 1.83 kg/ha. Conversely, India's pesticide usage stood at 0.44 kg/ha in 1990, declining to 0.37 kg/ha by 2021. Notably, in 2008, it plummeted to just 0.09 kg/ha, potentially attributed to the introduction of Bt cotton in 2002 and the subsequent reduction in pesticide use in the pesticide-consuming cotton crop.

3.2. Pesticide Consumption in India

Figure 3 shows the areas treated with different pesticides used in India from 2018–2019 (fiscal year from 1 April 2018 to 31 March 2019) to 2022–2023. The area treated with chemical pesticides ranged from 49% in 2021–2022 to 56% in 2018–19. In contrast, the area under exclusively bio-pesticide use ranged from 5 to 9% of the total cropped area, exhibiting a slight increasing trend over the years. The area treated with both chemical and bio-pesticides varied from 7% in years 2018–2019 and 2021–2022 to 18% in 2019–2020. Additionally, approximately 25% to 35% of the crop area was not treated with any pesticide.

India, a vast country with 23 states, has diverse agro-ecological conditions. Table 1 presents the state-wise pesticide consumption in India based on sales figures. Uttar Pradesh, the largest state in India in terms of agricultural land, has the highest pesticide consumption, followed by Maharashtra, Combined Andhra Pradesh, and Punjab (Table 1). Over the last decade, the total pesticide consumption has increased in Uttar Pradesh and Maharashtra, while it has slightly declined in Punjab, Rajasthan, Karnataka, Odisha, Bihar, Madhya Pradesh, Kerala, Gujarat, and Jammu and Kashmir. States like West Bengal and Kerala have seen a steep decline in the total consumption. Conversely, Chhattisgarh, Andhra Pradesh, and Tamil Nadu have shown a significant increase in total pesticide consumption. This indicates that pesticide use has not uniformly increased in all the states; in fact, some have even experienced declines. Overall, there has been a significant increase in pesticide usage,

with a 46% rise between the triennium ending (TE) 2007 and TE 2023. Overall, there was a significant rise of 46% in pesticide usage between the trienniums ending in 2007 and in 2023. Particularly noteworthy is the substantial surge in historically lower-pesticide-using states such as Jharkhand (an 833% increase), followed by Chhattisgarh (247%), Andhra Pradesh (265%), and Maharashtra (253%). Conversely, developed states already consuming higher quantities of pesticides per unit area, including Punjab, Haryana, West Bengal, Tamil Nadu, and West Bengal, have shown declining trends.

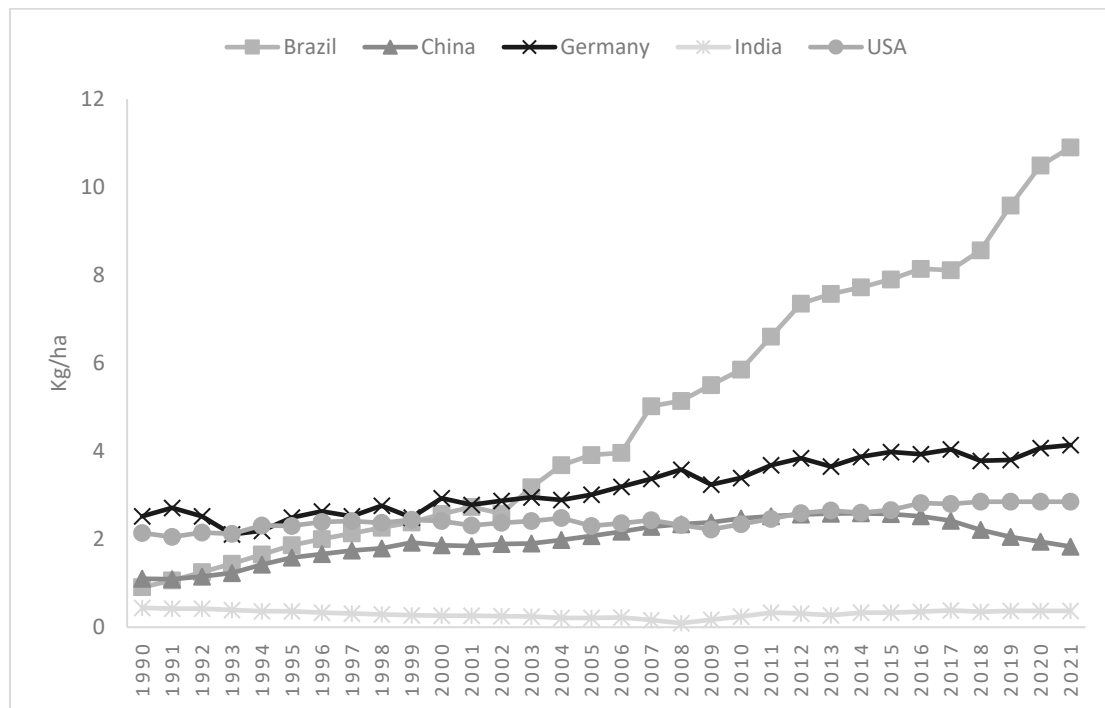


Figure 2. Pesticide use (kg/ha). Source: <https://www.fao.org/faostat/en/#data/RP>, accessed on 16 June 2024 [19].

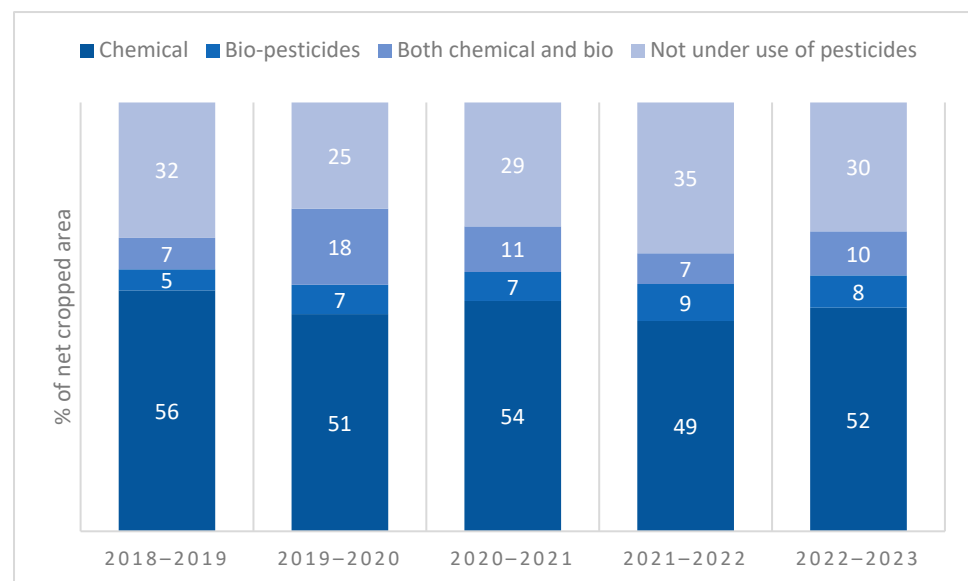


Figure 3. Area under pesticide use (% of net crop area). Source: imputed from reports of States/UTs Zonal Conferences on Inputs of Plant Protection for Kharif & Rabi Seasons, 2023–2024 [20].

Table 1. State-wise consumption (in tons) of pesticides (technical grade).

State	TE 2007	TE 2023	% Change
Uttar Pradesh	6980	11,690	67
Maharashtra	3140	11,077	253
Andhra Pradesh	1841	6715	265
Punjab	6162	5233	−15
Haryana	4560	4061	−11
West Bengal	4027	3527	−12
J&K	758	2607	244
Rajasthan	2068	2100	2
Karnataka	1733	1941	12
Tamil Nadu	2242	1879	−16
Gujarat	2757	1731	−37
Chhattisgarh	495	1718	247
Odisha	811	1249	54
Bihar	872	947	9
Jharkhand	74	687	833
Madhya Pradesh	831	648	−22
Kerala	492	540	10
Assam	167	449	170
Himachal Pradesh	301	269	−11
Uttarakhand	160	153	−4
All India	40,653	59,314	46

Note: Calculated from the annual reports of States/UTs Zonal Conferences on Inputs (Plant Protection) for Rabi & Kharif Seasons [20,21] and the Compendium of Environment Statistics India, 2011, Central Statistical Office, Ministry of Statistics and Programme Implementation, Government of India.

Another source of data on pesticides usage in India is the Cost of Cultivation Scheme, which does not provide the quantity of pesticides used but indicates the expenditure by farmers on pesticides. In 2011–2012, the share of pesticides in the cost of cultivation was 3% for cotton, 1.9% for paddy, 0.7% for wheat, and 0.3% for sugarcane, which are relatively low percentages. According to the Agricultural Input Survey data for 2011–2012, the highest percentage of area treated with pesticides was for cotton (66.70%), followed by pigeon pea (64.74%), jute (53.27%), and paddy (48.62%), with maize exhibiting 25.01% (All India Report on Input Survey 2011–2012). Between 1991–1992 and 2011–2012, there was a substantial escalation in the proportion of area treated with pesticides across all crops, except cotton. The decline in the pesticide-treated area for cotton began in 2002 with the introduction of Bt seeds. During the same period, the difference in the proportion of pesticide-treated areas under irrigated and unirrigated conditions narrowed, primarily due to the use of hybrids in rainfed areas that require effective pest management (All India Report on Input Survey 2011–2012) [17].

The recent input survey data [18] indicates a significant rise in pesticide use in some states and also highlights notable inter-state differences in pesticide use. The pesticide use in Jammu and Kashmir is significantly higher than that in other states due to the extensive area of fresh and dry fruit cultivation, especially apple plantations, which require the frequent application of pesticides (Figure 4). After Jammu and Kashmir, Punjab (1.3 kg/ha), Haryana (1.1 kg/ha), and Maharashtra (90.8 kg/ha) have higher concentrations of pesticide use and are considered affluent states. In contrast, pesticide use is lower in less developed states such as Madhya Pradesh, Rajasthan (0.1 kg/ha), Bihar (0.2 kg/ha), and Assam

(0.2 kg/ha). Cropping patterns also influence pesticide consumption; states with larger areas of cotton, chili, and vegetable cultivation consume more pesticides, while those with more area of cereal, pulse, and oilseed cultivation use fewer pesticides. The average consumption in India is 0.4 kg/ha.

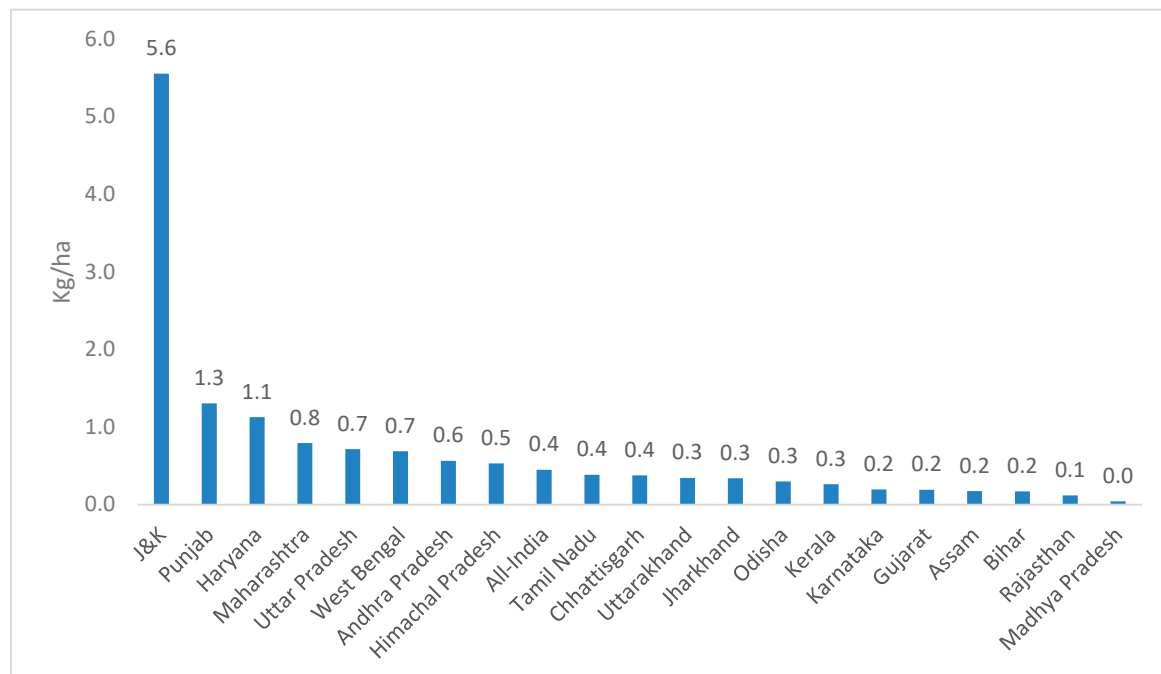


Figure 4. Pesticide use (kg/ha). Source: calculated from the All India Input Survey, 2016–2017 [18].

Figure 5 illustrates pesticide use in both irrigated and unirrigated agricultural lands. The difference in pesticide use between irrigated and unirrigated areas was much higher in Delhi, West Bengal, Haryana, Punjab, and Uttarakhand. In these states, farmers cultivate different crops in irrigated and unirrigated areas, contributing to the observed disparity. Conversely, in Maharashtra, Andhra Pradesh, and Madhya Pradesh, where similar crops are grown in both irrigated and unirrigated areas, the gap is smaller. This indicates that farmers use more pesticides under irrigated conditions, a finding corroborated by several studies showing that pesticides are more effective in irrigated lands.

Figure 6 illustrates the percentage change in pesticide usage between 2004–2005 and 2022–2023, based on the annual reports from the States/UTs Zonal Conferences on Inputs (Plant Protection) for Rabi & Kharif Seasons. The findings reveal significant increases in pesticide usage in Jammu and Kashmir (600%), Jharkhand (552%), Maharashtra (335%), and Chhattisgarh (258%). This surge can be attributed to two main factors: (i) a shift in cropping patterns toward fruits and vegetables, which typically require higher pesticide application per unit area, and (ii) a lower baseline, as these states had comparatively low pesticide consumption during the base year. Conversely, states with more advanced agricultural sectors, such as Gujarat, Tamil Nadu, Punjab, Madhya Pradesh, Haryana, Himachal Pradesh, and West Bengal, displayed negative growth rates, indicating a decline in pesticide consumption compared to the higher levels observed in 2004–2005.

It is pertinent to note that there are discrepancies in comparing the per-hectare use of pesticides between the All India Report on Input Surveys and the annual reports of the States/UTs Zonal Conferences on Inputs (Plant Protection) for Rabi & Kharif Seasons. The former is based on farmers' actual use surveys, while the latter is based on state-level computations derived from retail sales of pesticides. Therefore, there is a need to improve the methodology for capturing precise data on pesticide consumption at the farmer level in the Zonal Conferences on Inputs. The Zonal Conferences are conducted annually,

while the input surveys are conducted every five years with a more precise representative sampling framework.

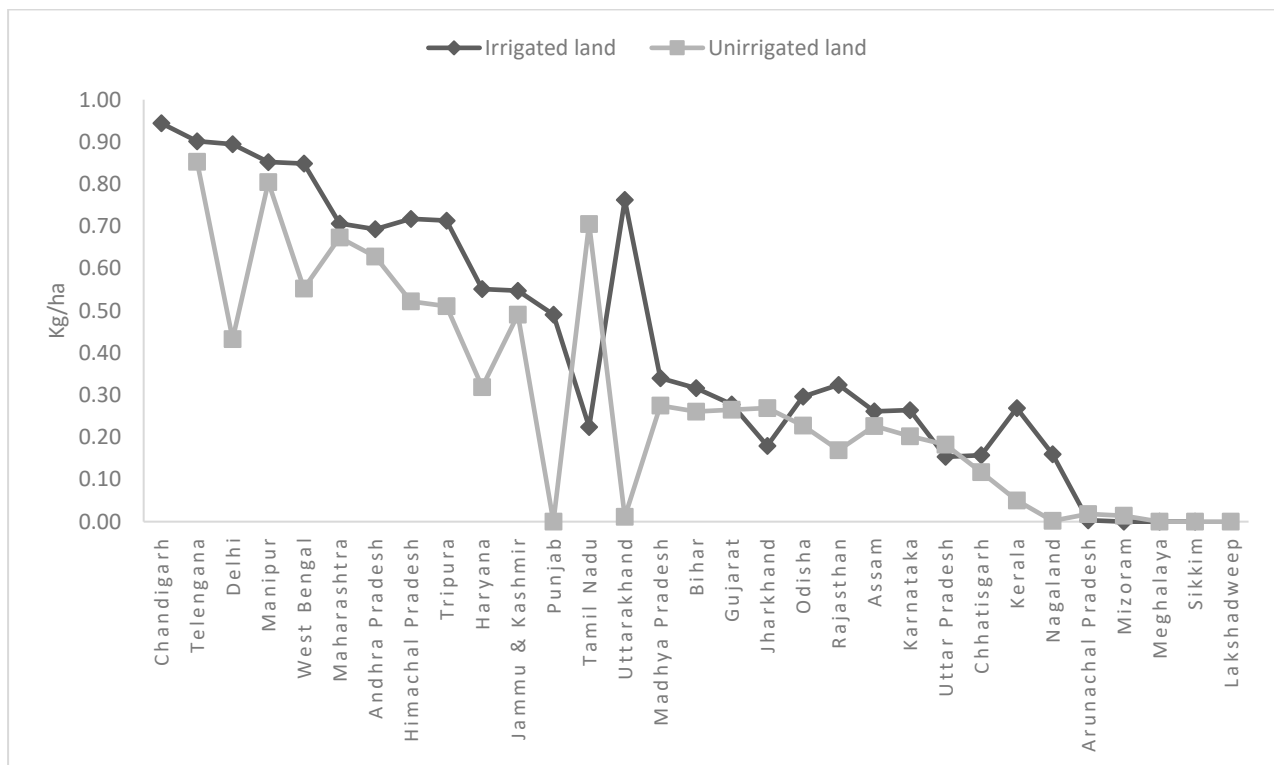


Figure 5. State-wise pesticide use (kg/ha). Source: calculated from the All India Input Survey, 2016–2017 [18].

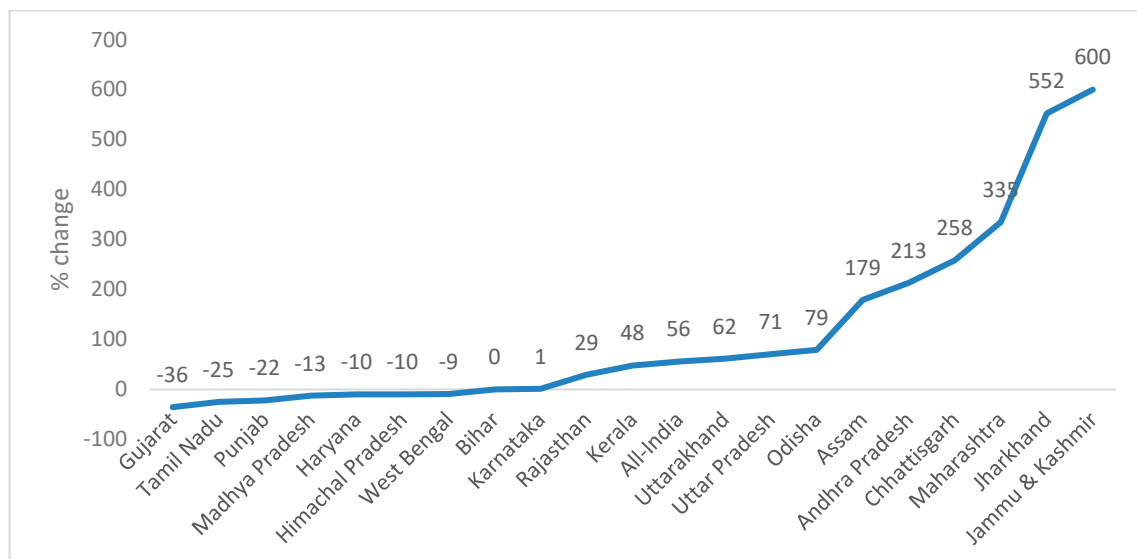


Figure 6. Per cent change in pesticide use between 2004–2005 and 2022–2023. Source: Note: calculated from annual reports of the States/UTs Zonal Conferences on Inputs (Plant Protection) for Rabi & Kharif Seasons [20,21] and the Compendium of Environment Statistics India, 2011, Central Statistical Office, Ministry of Statistics and Programme Implementation, Government of India.

Table 2 provides information on pesticide use across different land size categories under both irrigated and unirrigated conditions. Approximately 39% of farmers using irrigated condition treated their crops with pesticides, compared to 36% of farmers em-

ploying unirrigated conditions. This supports the general understanding that irrigated farming tend to use more inputs, such as pesticides, to boost productivity and capitalize on favorable conditions. In contrast, unirrigated farming may use fewer pesticides due to economic constraints or the reduced effectiveness of pesticides under the farming conditions. Notably, large farmers employing irrigated conditions use more pesticides, whereas large farmers employing unirrigated conditions use fewer. Among farmers using unirrigated conditions, marginal and small farmers use more pesticides, possibly due to their intensive farming practices aimed at maximizing income from limited land.

Table 2. Area treated with pesticides in irrigated and unirrigated areas by land size category.

Land Size Category	Area Treated with Pesticides	
	Irrigated Area (%)	Unirrigated Area (%)
Marginal (<1 ha)	37	39
Small (1–2 ha)	39	40
Semi-medium (2–4 ha)	39	38
Medium (4–10 ha)	39	31
Large (>10 ha)	42	24
All	39	36

Source: calculated from the All India Input Survey, 2016–2017 [18].

Figure 7 illustrates the distribution of pesticide-treated areas across various crops, revealing substantial disparities in pesticide application. Although the sample size is small, input survey data indicates that pesticide consumption is 100% for apples and some commercially important fruits and vegetables. Among major crop categories, jute exhibits the highest proportion of treated area at 67%, followed by cotton at 57%. Pigeon pea follows closely with 52%. Conversely, crops like pearl millet and wheat have lower pesticide-treated areas, at 22% and 34%, respectively, while mustard and paddy both stand at 35%.

To address these disparities and promote more sustainable pesticide use, policymakers could consider several strategies. One approach involves targeted education and outreach programs for farmers, emphasizing integrated pest management practices that incentivize optimal pesticide use. Additionally, implementing stricter regulations on pesticide use, particularly for crops with high treatment rates like chickpea and cotton, could encourage more judicious and responsible pesticide application. By adopting these policy options, stakeholders can work toward a more balanced and environmentally friendly approach to pest management in agriculture.

Figure 8 depicts the use of pesticides across different crops in both irrigated and unirrigated areas. Interestingly, in crops such as jute, mustard, maize, paddy, wheat, and pearl millet, more pesticides were used in irrigated areas. Conversely, for crops such as cotton, pigeon pea, gram, sorghum, groundnut, and sugarcane, more fertilizers were used in unirrigated lands. This indicates that while the general trend is toward higher pesticide use in irrigated lands, certain crops exhibit the opposite pattern.

Table 3 presents the expenditure on pesticides per unit area, categorized by crop and state. The highest expenditure per hectare is observed in Haryana, Andhra Pradesh, Punjab, Telangana, and Karnataka for most crops. In contrast, the lowest expenditure is seen in Rajasthan, Jharkhand, Assam, and Bihar. Among the crops, cotton has the highest average pesticide expenditure per hectare, followed by onion, pigeon pea, and paddy. The lowest expenditures are noted in sesame, maize, and sorghum. Overall, the data indicates that farmers in agriculturally affluent states and those growing commercial crops use more pesticides, whereas farmers in less agriculturally developed states and those cultivating millet and oilseed use fewer pesticides.

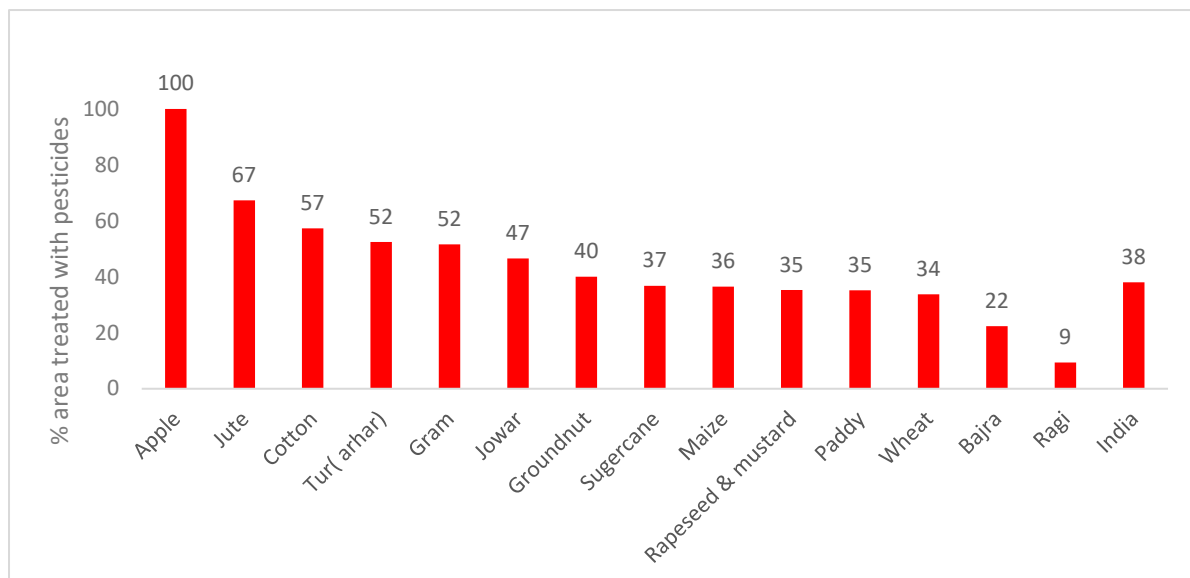


Figure 7. Area treated with pesticides (%) for major crops. Note: calculated from the All India Input Survey, 2016–2017 [18].

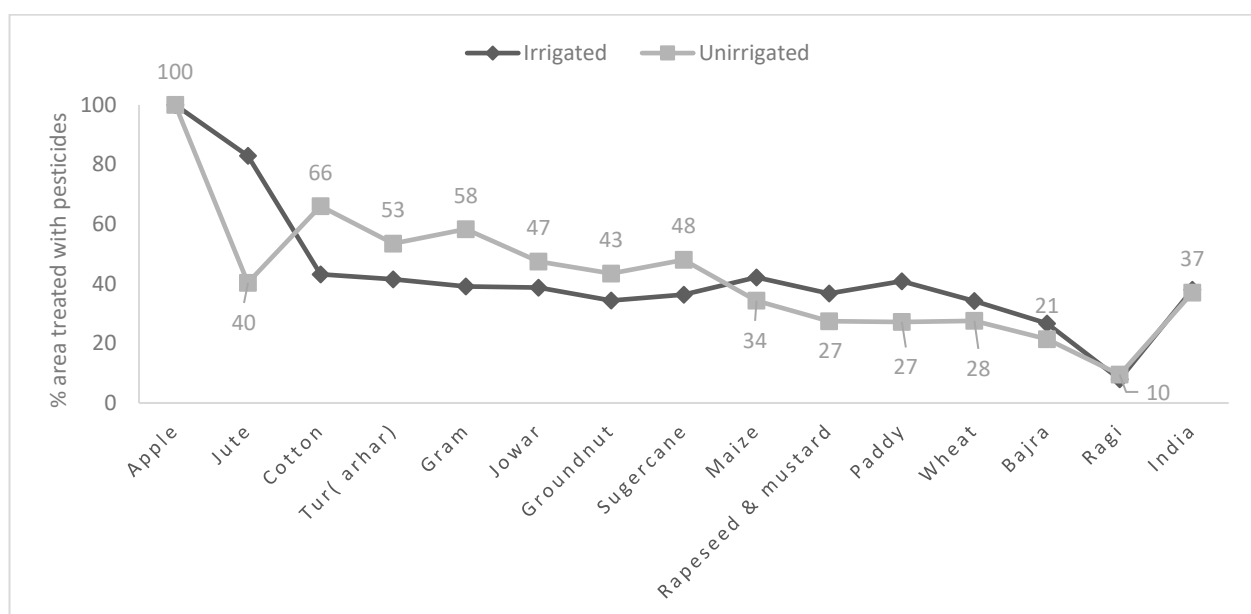


Figure 8. Area treated with pesticides in irrigated and unirrigated areas (%) in 2016–2017. Note: calculated from the All India Input Survey, 2016–2017 [18].

3.3. Use of Bio-Pesticides

Figure 9 illustrates the aggregate consumption of chemical and bio-pesticides across various crop types. Cereals emerge as the highest consumers of chemical pesticides, followed sequentially by pulses, cash crops, oilseeds, and vegetables. Even bio-pesticide usage is most prominent in cereals, followed by oilseeds, vegetables, and cash crops. These findings underscore the critical need for targeted policy measures to foster sustainable pesticide management. To address this, policymakers could prioritize research and development efforts to develop bio-pesticides specifically suited to the pest challenges in cereal crops. Additionally, extension services and farmer training programs should focus on promoting integrated pest management (IPM) strategies that advocate for the judicious use of both chemical and bio-pesticides to minimize environmental impacts while ensuring effective pest control. Furthermore, encouraging the adoption of pest-resistant crop vari-

eties through subsidies or support programs could significantly reduce the dependence on chemical pesticides, especially in crops with high pesticide consumption such as cereals and pulses.

Table 3. Crop-wise and state-wise expenditure on pesticides (Rs/ha).

State	Crops									
	Cotton	Onion	Pigeon Pea	Paddy	Moong	Soybean	Groundnut	Jowar	Maize	Sesame
Haryana	3316			6960						
Andhra Pradesh	6194	6041	2662	6053	7840		1896	5687	5593	413
Punjab	6753			5841	8740				2200	
Telangana	4801		3793	5248	1576	1782	3971		3500	
Karnataka	3699	2366	2427	2399	891	1698	368	453	837	
Madhya Pradesh	2315	1944	1070	2282		2519			2236	667
Tamil Nadu	3066	3021		2165	1349		813	169	2105	
Kerala				2005			101			
Chhattisgarh				1568						
Himachal Pradesh				1216					212	
Maharashtra	4478	5510	5235	1158	873	3026	297	15	1702	
Gujarat	3703	4237	2282	1107	918	901	4177		843	2010
Odisha	1905		3	1026	113				892	
Uttar Pradesh				972	93		156		174	
West Bengal				972	110		3146			951
Bihar				167					8	
Assam				24						
Jharkhand										
Rajasthan	3642	1115			422	2306	1414		194	
Average	3988	3462	2496	2421	2084	2039	1634	1581	1577	1010

Note: calculated from Cost of Cultivation Scheme data (2021–2022).

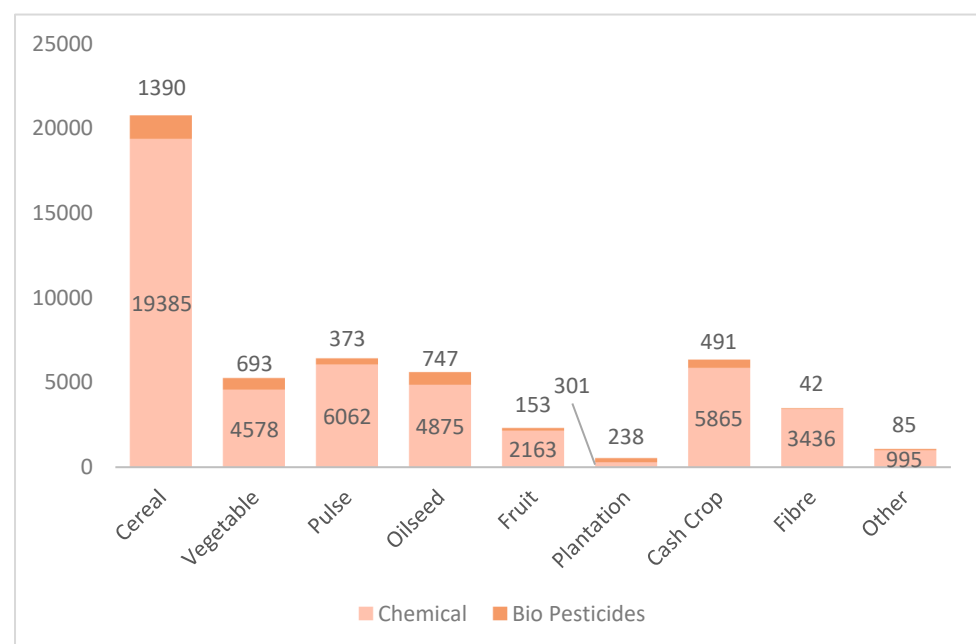


Figure 9. Use of chemical and bio-pesticides in different crops in '000 tons in 2023–2024. Note: calculated from the States/UTs Zonal Conferences on Inputs of Plant Protection for Kharif & Rabi Seasons [18].

Figure 10 delineates the proportion of chemical and bio-chemical pesticides used across various crop groups. Notably, the share of chemical pesticides is disproportionately high, with cotton showing a 99% usage rate and oilseeds and vegetables at 87%. Conversely, the share of bio-pesticides is significantly higher in plantation crops (44%), followed by oilseeds and vegetables. To address this imbalance, policymakers could implement measures to incentivize the production and adoption of bio-pesticides through tax incentives or

subsidies for bio-pesticide manufacturers and farmers. Additionally, stricter regulations on chemical pesticide usage could be enforced, accompanied by educational campaigns to raise awareness about the benefits of transitioning toward more sustainable pest management practices. By implementing these policy measures, stakeholders can work toward a more balanced and environmentally friendly approach to pesticide use in agriculture, ensuring the long-term sustainability of food production systems.

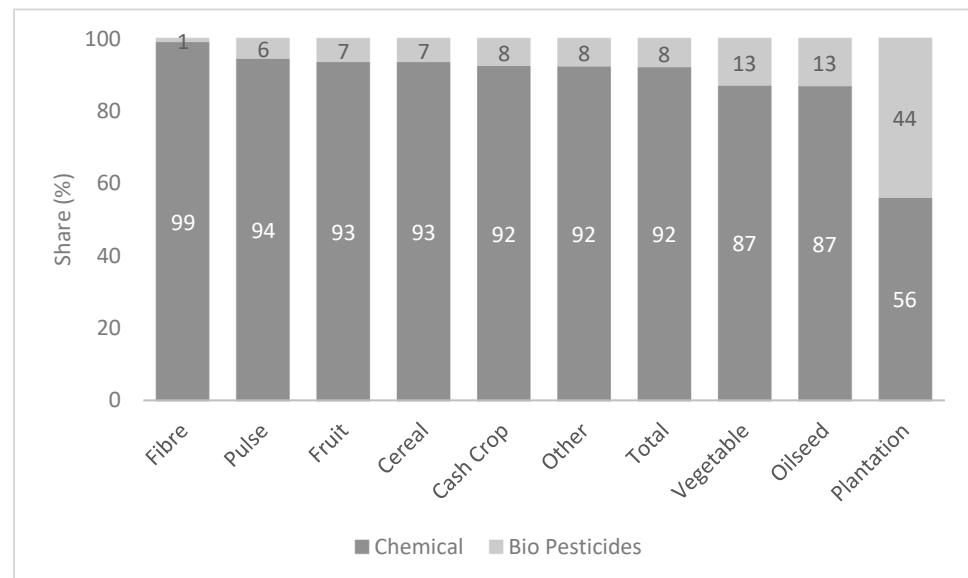


Figure 10. Share (%) of chemical and bio-pesticides use in different crops in 2022–2023. Source: Note: calculated from the States/UTs Zonal Conferences on Inputs of Plant Protection for Kharif & Rabi Seasons [18].

3.4. Use of Integrated Pest Management (IPM)

It is important to recognize that pesticides are just one component of the pest management strategies employed by farmers. Ideally, farmers use a combination of traditional (agro-ecological and mechanical) strategies and chemical and bio-pesticides to control pests (Table 4). Among the surveyed farmers, approximately 72% reported adopting pest control measures. Of these, 39% opted for chemical methods, 24% for agro-ecological methods, 9% for mechanical methods, and 3% for biological methods. Additionally, 18% used alternative methods, while about 28% did not implement any pest control measures.

- **Agro-ecological methods:** These methods emphasize the integration of natural processes and biodiversity to sustainably manage pests. They include crop rotation, polyculture, and the use of natural predators to reduce pest populations. By fostering a diverse ecosystem, beneficial insects and organisms thrive, which naturally keeps pest numbers in check. Additionally, practices such as habitat management and using pest-resistant crop varieties minimize the need for chemical pesticides, promoting environmental health and reducing the risk of pest resistance.
- **Mechanical methods:** These involve physical techniques and devices to manage and reduce pest populations, such as hand picking pests, using traps and barriers, and employing machinery like plows and cultivators to disrupt pest habitats. Techniques such as mulching and soil solarization can also create unfavorable conditions for pests. Mechanical control minimizes the use of chemical pesticides, thereby reducing environmental impact and health risks to humans and non-target species. These methods offer immediate and effective solutions, especially in smaller-scale or organic farming operations.
- **Biological methods:** These involve using living organisms to suppress pest populations through natural predation, parasitism, and competition. They include introducing

or conserving beneficial insects like ladybugs and predatory beetles, which feed on pests such as aphids and caterpillars, and using parasitic wasps that lay eggs inside pest larvae. Microbial agents like *Bacillus thuringiensis* (Bt), a bacterium that produces toxins harmful to specific insects, provide targeted pest management. Biological control methods are sustainable and environmentally friendly, reducing reliance on chemical pesticides and fostering ecological balance in agricultural systems.

In terms of regional adoption, the following findings are observed:

- In Odisha, 100% of households adopted some type of pest control measures (Table 4).
- In Haryana, Punjab, and Gujarat, about 99% of farmers implemented pest control measures.
- Andhra Pradesh had a 96% adoption rate, West Bengal 94%, and Jammu 93%.
- Conversely, Uttarakhand had only 29% adoption, Uttar Pradesh 36%, and Jharkhand 58%.

Table 4. Adoption of different types of pest control measures in different states.

State	Households Adopting Pest Control Measures (%)	Chemical Control (%)	Agro-Economic and Cultural Practices (%)	Mechanical Control (%)	Biological Control (%)	Other (%)	No Efforts (%)
Telangana	92	88	14	15	6	0	8
West Bengal	94	83	1	2	2	11	6
Maharashtra	89	69	65	37	5	0	11
Andhra Pradesh	96	61	39	19	4	19	4
Haryana	99	58	8	31	2	17	1
Himachal Pradesh	74	57	5	0	1	16	26
Punjab	99	56	31	31	0	0	1
Jammu	93	53	25	7	8	27	7
Tamil Nadu	90	46	41	9	0	23	10
Madhya Pradesh	74	41	9	1	3	25	26
Gujarat	99	30	83	26	1	10	1
Odisha	100	30	4	0	3	67	0
Bihar	79	29	4	2	1	45	21
Rajasthan	75	28	62	4	0	1	25
Jharkhand	58	28	21	1	1	13	42
Uttarakhand	29	28	0	0	0	1	70
Assam	66	24	15	8	3	20	34
Karnataka	69	22	19	5	2	32	31
Uttar Pradesh	36	18	0	0	7	12	64
Chhattisgarh	62	17	50	2	1	3	38
Kerala	25	3	9	1	1	12	75
India	72	39	24	9	3	18	28

Source: Agricultural Input Survey, 2016–2017 [18].

Regarding chemical methods, Telangana had the highest adoption rate at 88%, followed by West Bengal (83%), Maharashtra (69%), Andhra Pradesh (61%), and Haryana (58%). However, states like Kerala (9%), Chhattisgarh (17%), Uttar Pradesh (18%), and Karnataka (22%) had lower adoption rates.

In terms of agro-economic and cultural practices, the adoption rates varied significantly across different states. In Gujarat, 83% of farmers employed these methods, followed by 65% in Maharashtra, 62% in Rajasthan, 50% in Chhattisgarh, and 41% in Tamil Nadu. Conversely, Uttarakhand and Uttar Pradesh reported zero adoption rates. Other states with adoption rates below 10% included West Bengal, Bihar, Odisha, Himachal Pradesh, Haryana, Kerala, and Madhya Pradesh.

Mechanical methods were predominantly utilized by farmers in Maharashtra (37%), followed by Punjab and Haryana (31%) and Gujarat (26%). No adoption rates of mechanical methods were reported in Uttarakhand, Uttar Pradesh, Odisha, or Himachal Pradesh.

Biological control methods were most commonly used in Jammu (8%), followed by Uttar Pradesh (7%) and Telangana (6%). In contrast, adoption rates were nearly zero in Uttarakhand, Rajasthan, Tamil Nadu, and Punjab.

In terms of farmers not engaging in any pest control efforts, the highest rates were observed in Kerala (75%), Uttarakhand (70%), Uttar Pradesh (64%), and Jharkhand (42%). In comparison, Odisha, Punjab, Gujarat, and Haryana had significantly lower rates of non-adoption.

3.5. Composition of Pesticide Production in India

In India, pesticide production is primarily dominated by insecticides and fungicides, followed by herbicides and rodenticides (Figure 11). The proportion of insecticides has shown fluctuations, decreasing from over 42% in 2018–2019 to 31% in 2021–2022 before rising again to 42% in 2022–2023. Fungicides and herbicides exhibited an upward trend until 2021–2022, followed by a decline in 2022–2023. Conversely, the share of rodenticides has been steadily increasing over time. The significant growth in fungicide use is largely attributed to their application in fruit and vegetable crops. Some of the major pesticides produced in India include mancozeb, 2-4-D, acephate, and profenofos.

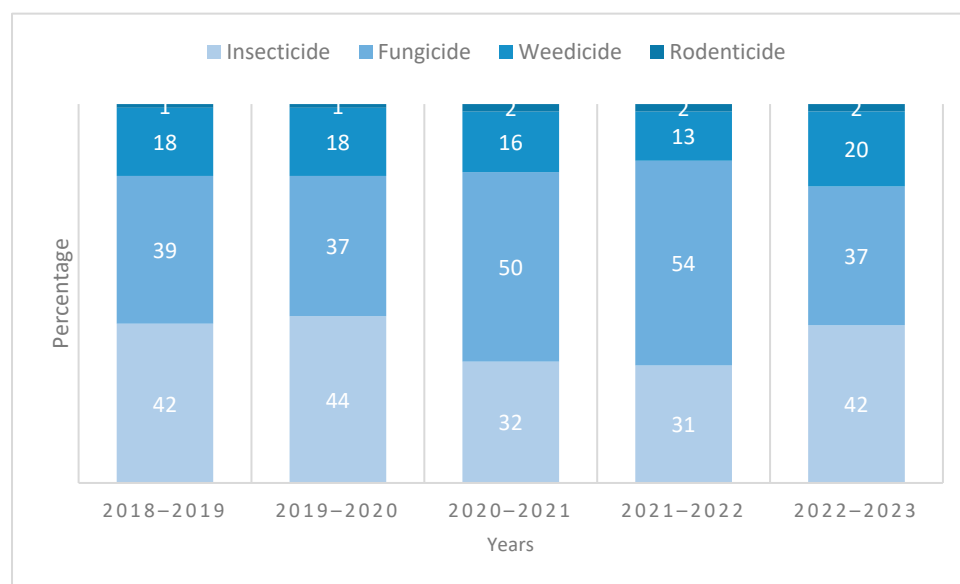


Figure 11. Share (%) of different pesticides produced in India. Source: <https://ppqs.gov.in>, accessed on 1 January 2024 [22].

3.6. Pesticide Production, Imports, Exports, and Consumption in India

Pesticides, including insecticides, rodenticides, fungicides, herbicides, anti-sprouting products, plant growth regulators, and disinfectants, are classified under HS code 3808. In India, insecticides dominate the overall pesticide usage. Table 5 provides comprehensive details on total pesticide production, imports, consumption, and exports. Over the years, there has been a noticeable upward trend in pesticide consumption. For instance, consumption increased from 40,000 tons in 2005–2006 to 63,000 tons by 2021–2022. This surge is partly attributed to the expansion of input-intensive agriculture, as noted by the Directorate of Plant Protection, Quarantine, and Storage.

In the fiscal year 2022–2023, India recorded an export value of 0.63 million tons of pesticides amounting to Rs 431.64 billion. The recent increase in pesticide usage is primarily driven by higher herbicide utilization, prompted by the rising costs of manual weed control due to increasing agricultural wages. India has emerged as a leading global exporter of pesticides, second only to China. Domestic production surpasses consumption by a considerable margin, leading to a surge in exports.

It is noteworthy that the sum of pesticide exports and domestic consumption exceeds the combined total of domestic production and imports. This gap arises from the processing and repackaging within the country. Several Indian companies often import raw pesticides or bulk formulations, process them (e.g., dilution, formulation adjustments), repackage them, and then export the finished products. This processing adds volume to the imported goods, allowing the country to export more than it produces domestically.

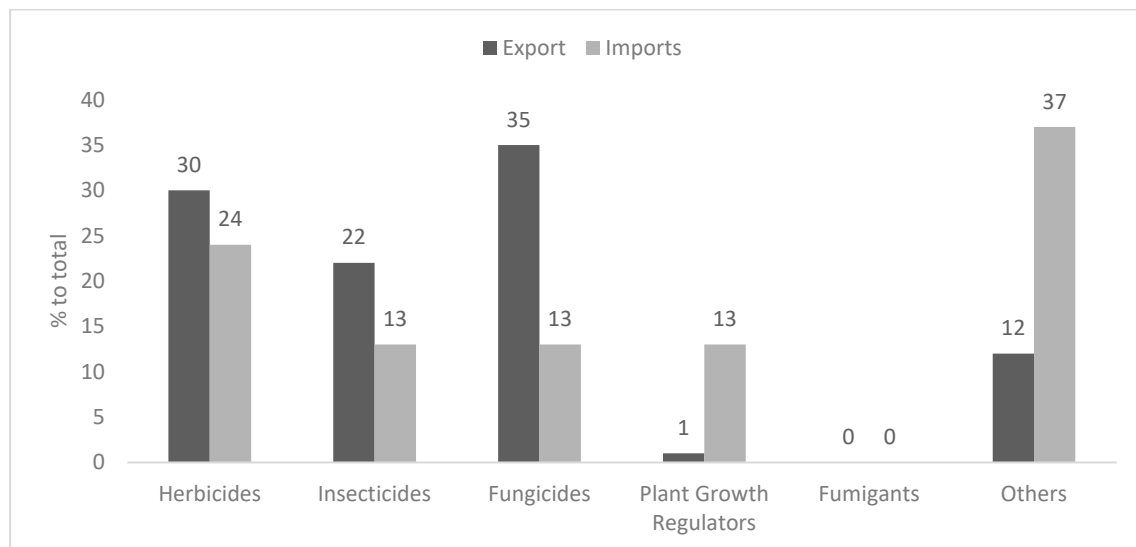
Table 5. Trends in production, import, export, and consumption of pesticides (technical grade) in India (in 1000 tons).

Year	Production	Import	Total	Consumption	Export
2005–2006	82	19	101	40	91
2006–2007	85	28	113	42	108
2007–2008	80	29	109	44	96
2008–2009	85	18	104	44	185
2009–2010	82	22	104	42	126
2018–2019	217	117	333	60	461
2019–2020	192	107	298	62	452
2020–2021	255	157	412	62	533
2021–2022	298	134	432	63	648
2022–2023	258	134	392	52	630

Source: States/UTs Zonal Conferences on Inputs (Plant Protection) for Rabi & Kharif Seasons [22].

3.7. Trade in Pesticides

In the year 2023, pesticide exports from India surged to a record high of US\$ 5.4 billion (INR 45,117 crore). Fungicides represented the largest share of exports by quantity, accounting for 35%, followed by herbicides at 30% (Figure 12). Among the imported pesticide categories—herbicides, insecticides, and fungicides—herbicides dominated in terms of volume.

**Figure 12.** Export and import (% to total) of major agro-chemicals by India, 2022–2023. Source: Computed from the States/UTs Zonal Conferences on Inputs (Plant Protection) for Rabi & Kharif Seasons [22].

According to a 2021 report from the Federation of Indian Chambers of Commerce and Industry (FICCI), the top five pesticides exported from India were mancozeb, cypermethrin, 2,4-D, acephate, and chlorpyrifos. In contrast, glyphosate and atrazine were among the major imported products. Interpreting trade data requires caution, as firms trade both formulations and technical-grade pesticides. Indian firms primarily import technical-grade or patented formulations, while their exports are mainly in the form of formulations.

Brazil, Bangladesh, Nigeria, the United Arab Emirates, and Argentina emerge as significant destinations for India's pesticide exports (Table 6). Conversely, China and Israel are major sources of pesticide imports for India, from which technical-grade pesticides are

imported, according to the data provided by the Ministry of Chemicals and Petrochemicals, Government of India.

Table 6. Major export and import destinations for agro-chemicals in 2023–2024 (tons).

	Country	Insecticide	Fungicide	Herbicide
Export	Brazil	50,327	58,046	19,545
	Bangladesh	6856	30,272	0
	Nigeria	4550	0	0
	Arab Emirates	0	16,072	0
	Argentina	0	0	7508
	USA	0	0	30,589
Import	China	13,834	5904	35,314
	Israel	1105	0	4665
	Japan	796	0	0
	Thailand	0	1813	0
	Belgium	0	1763	0
	USA	0	0	12,922

Source: States/UTs Zonal Conferences on Inputs (Plant Protection) for Rabi & Kharif Seasons [22].

Glyphosate and atrazine are the major pesticides imported by quantity, with China as the primary source. Domestic manufacturers import technical-grade glyphosate and then produce formulations for the domestic market. Additionally, large companies export glyphosate to countries in Latin America and Africa. Atrazine is imported by various firms, both domestic and multinational, and is predominantly converted into formulations for the domestic market. Acephate is as a major pesticide exported to Brazil, with one large company accounting for approximately 80% of the total quantity. Similarly, another prominent company exports roughly 75% of the total quantity of mancozeb to various countries in Africa and Latin America.

3.8. Market Share of Different Pesticide Categories in India

The market share of the top 10 pesticide products is important for governments, policy makers, and farmers. For governments and policy makers, understanding which products dominate the market helps in developing equitable regulations, ensuring healthy competition, and promoting safe farming practices. This knowledge prevents market monopolization and informs policy decisions that protect both farmers and consumers. For farmers, the prevalence of certain pesticide products indicates their reliability and effectiveness, in effect guiding their purchasing decisions based on collective trust and efficacy.

This section analyzes the market share of different pesticide formulations and trends in market concertation. Table 7 illustrates the market share of various pesticide formulations in India. In the category of insecticides, the top 10 formulations collectively hold a market share of 59%. Leading formulations include chlorpyrifos (14%), malathion (7%), and quinalphos (6%). For fungicides, the top ten formulations command an 86% market share. Sulfur is the predominant formulation, occupying 40% of the market, followed by mancozeb (22%) and carbendazim (7%). Among weedicides, the top ten formulations hold an 87% market share. The primary formulations in this category are glyphosate (15%), 2,4-D amine salt (15%), and pretilachlor (12%).

Understanding these market shares helps governments support research and provide financial assistance for the best and safest pesticides, ensuring farmers have access to reliable pest control options.

Table 8 outlines the market share of the top ten rodenticides, plant growth regulators, and bio-pesticides in India. Among rodenticides, the top 10 formulations collectively

hold a market share of 100%. The primary formulations include zinc phosphide (35%), aluminum phosphide (33%), methyl bromide (13%), and bromadiolone (10%). For plant growth regulators, the top ten formulations also command a combined market share of 100%. Notable formulations include paclobutrazol (19%), alpha naphthyl acetic acid (17%), and validamycin (16%). Among bio-pesticides, the top 10 formulations have a 100% market share. Key formulations include *Pseudomonas fluorescens* (16%), *Tricoderma* spp. (15%), neem-based insecticides (12%), and *Metarrhizium anisopliae* (12%).

Table 7. Top ten pesticide formulations and their market share (insecticides, fungicides, and weedicides).

Insecticide	% Share	Fungicide	% Share	Weedicide	% Share
Chlorpyrifos	14	Sulfur	40	Glyphosate	15
Malathion	7	Mancozeb	22	2,4-D Amine salt	15
Quinalphos	6	Carbendazim	7	Pretilachlor	12
Cypermethrin	5	Propineb	3	Butachlor	10
Monocrotophos	5	Ziram	3	2,4-D Dichlorophenoxy	10
Fipronil	5	Copper oxychloride	3	Atrazine	9
Profenophos	5	Captan	3	Pendimethalin	5
Fenvalerate	5	Zineb	2	Isoproturon	4
Acephate	4	Dodine	2	Chlodinafop-propargyl	3
Dimethoate	4	Hexaconazole	2	Anilophos	3
Share of top 10	59		86		87

Source: <https://ppqs.gov.in>, accessed on 1 January 2024 [22].

Table 8. Top ten pesticide formulations and their market share (rodenticides, plant growth regulators, bio-pesticides).

Rodenticide	% Share	Plant Growth Regulator	% Share	Bio-Pesticide	% Share
Zinc phosphide	35	Paclobutrazol	19	<i>Pseudomonas fluorescens</i>	16
Aluminum phosphide	33	Alpha naphthyl acetic acid	17	<i>Tricoderma</i> spp.	15
Methyl bromide	13	Validamycin	16	Neem-based insecticides	12
Bromadiolone	10	Triacantanol	15	<i>Metarrhizium anisopliae</i>	12
Ethylene dibromide	4	Chlormequat chloride	12	<i>Tricoderma viride</i>	12
Barium carbonate	1	Gibberellic acid	11	<i>Metarhizium rileyi</i>	11
EDCT mixture	1	Growth promoters	9	<i>Beauveria bassiana</i>	8
Coumachlor	1	Sodium paranitro phenolate	7	<i>Verticillium lecanii</i>	6
Warfarin	0			Azadirachin	5
				NPV (H)	4
Share of top 10	100		100		100

Source: <https://ppqs.gov.in>, accessed on 1 January 2024 [22].

3.9. Distribution of Sales and Reach to Consumers

India boasts an extensive network of sales points (input dealers) spread across its 7000 blocks (sub-districts), as well as in large villages and village clusters. Nationwide, the concentration of sale points averages two per 1000 ha (Figure 13).

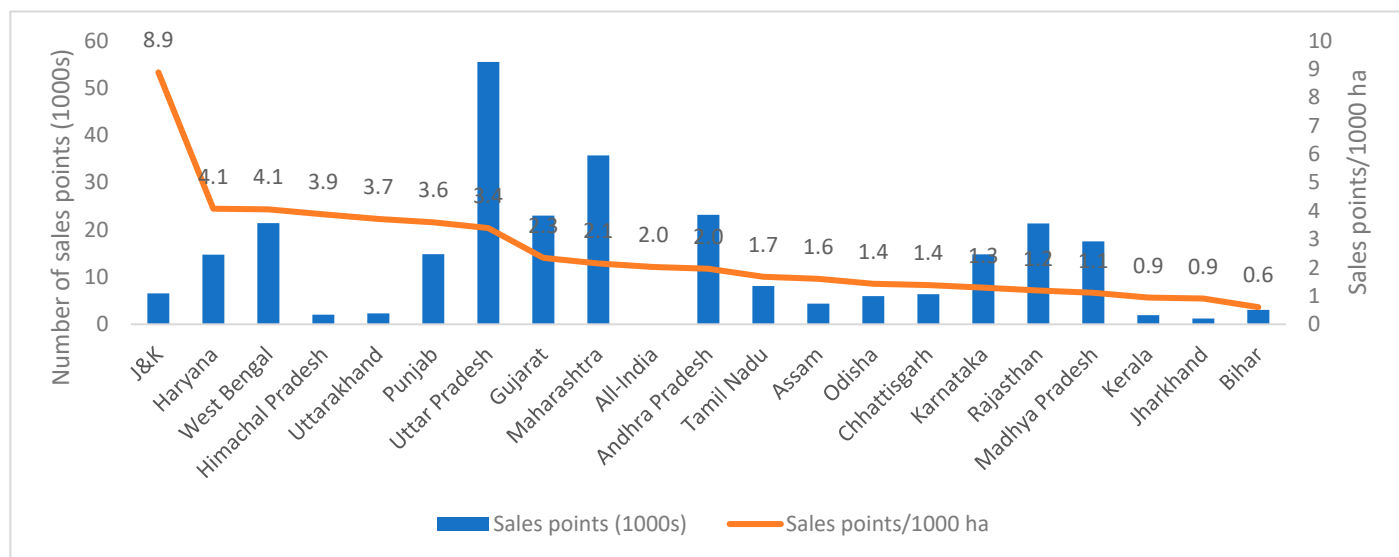


Figure 13. Distribution of sales points of pesticides. Source: <https://ppqs.gov.in>, accessed on 1 January 2024 [22].

- Regions with high concentrations: Jammu and Kashmir has the highest concentration with 8.9 sales points per 1000 hectares, followed by Haryana (4.1), West Bengal (4.1), Himachal Pradesh (3.9), Punjab (3.6), and Uttar Pradesh (3.4).
- Regions with low concentrations: Bihar has the lowest concentration with 0.6 sales points per 1000 hectares, followed by Jharkhand (0.9), Kerala (0.9), Madhya Pradesh (1.1), and Rajasthan (1.2).

These regional disparities in the concentration of sales points highlight the need for policies aimed at reducing such disparities.

4. Regulation, Registration, and Quality Control

The Insecticides Act of 1968 [23] and Insecticides Rules (1971) [24] serve as the primary regulatory frameworks governing the import, registration, production, sale, transport, distribution, use, and disposal of pesticides in India. These regulations aim to protect crops from pests and diseases while mitigating risks to humans and animals. All pesticides must undergo a rigorous registration process with the Central Insecticides Board and the Registration Committee (CIB&RC) prior to production or sale. Applicants seeking approval for manufacturing or importing pesticides must submit comprehensive data on chemical composition, toxicity, and bioefficacy to the CIB&RC [25]. Additionally, published reports from reputable R&D organizations, particularly regarding the bioefficacy of pesticides, are considered valid data sources.

Upon validation, the CIB&RC issues a registration number and certificate. As of March 2024, 339 pesticides and 946 pesticide formulations are registered under the Insecticides Act of 1968 [26]. Various regulations and procedures govern the testing of pesticides at different stages. The Central Insecticide Laboratory (CIL) is responsible for testing referral samples submitted by any officer or agency of the Central or State Government, while the State Pesticide Testing Laboratories (SPTLs) primarily conduct quality control testing on samples from manufacturing facilities and points of sale.

As of March 2024, 29 pesticides are banned for use in India, while 5 pesticides are prohibited for use but have been granted permission for manufacturing. Additionally, 16 pesticides have been refused permission for use in the country [27–31]. The entire list of banned pesticides is available at https://ppqs.gov.in/sites/default/files/list_of_pesticides_which_are_banned_refused_registration_and_restricted_in_use_0.pdf, accessed on 1 January 2024.

The industry grapples with several challenges, including stringent global environmental regulations, limited investment in research and development by domestic manufacturers due to high costs the need for innovation and product diversification, insufficient awareness among farmers regarding the safe use of pesticides, lengthy product development timelines, and the assurance of product quality [31]. Safeguarding farmers from substandard products is crucial, and initiatives to promote safe pesticide usage and mitigate health and environmental risks should be prioritized. It is crucial to ensure farmers' protection against substandard products by enforcing quality inspections at points of sale and through consumer forums. The government must take proactive steps to eliminate counterfeit pesticides by strengthening the regulatory framework, improving quality control, enhancing farmer education, leveraging technology, and fostering multi-stakeholder collaboration.

Over the past decade, the proportion of extremely and highly hazardous pesticides (classified by the WHO) in total production has declined. In 2018–2019, extremely hazardous pesticides (WHO Class Ia) accounted for 6.62% of total production, highly hazardous pesticides (WHO Class Ib) 4.81%, moderately hazardous pesticides (WHO Class II) 38.26%, slightly hazardous pesticides (WHO Class III) 6.79%, and those unlikely to present acute hazards (WHO Class U) 39.36%.

Bio-pesticides, which constitute approximately 9% of India's pesticide market, offer a promising avenue for reducing crop losses and minimizing environmental impacts [32]. Currently, 71 bio-insecticides and 31 bio-fungicides are registered under the Insecticides Act of 1968. Bio-pesticide consumption has increased significantly, from 219 tons in 1996–1997 to approximately 7200 metric tons in 2022–2023. Research indicates that integrating bio-pesticides into pest management strategies can substantially reduce pesticide use by up to 66% in cotton and 45% in cabbage [33,34].

However, the development of the bio-pesticide market remains slow, requiring specialized facilities and skills for storage and formulation. Fiscal incentives may be necessary to encourage the production and use of bio-control agents.

4.1. Labeling of Pesticide Products

The proper labeling of pesticides is essential to ensure their safe use. The failure to display mandatory warnings or cautionary messages on pesticide labels is considered "misbranding" under Sec 3 (k) (iii) and is subject to prosecution under Sec 29 (1) (a) of the Insecticides Act of 1968 [23,24].

Labeling requirements (Table 9):

- Labels must prominently feature a diamond-shaped square occupying at least one-sixteenth of the total label area.
- The upper portion of the square must contain symbols and signal words indicating toxicity levels:
 - (i). Category I (extremely toxic): Skull and crossbones symbol and "POISON" in red, with warnings "KEEP OUT OF THE REACH OF CHILDREN" and "IF SWALLOWED, OR IF SYMPTOMS OF POISONING OCCUR, CALL PHYSICIAN IMMEDIATELY".
 - (ii). Category II (highly toxic): "POISON" in red and "KEEP OUT OF THE REACH OF CHILDREN".
 - (iii). Category III (moderately toxic): "DANGER" and "KEEP OUT OF THE REACH OF CHILDREN".
 - (iv). Category IV (slightly toxic): "CAUTION".

The lower portion of the square should contain the color specified in column (4) of Table 9, depending on the classification of the insecticide specified in the corresponding entry in column (1) of the table.

In addition to the above precautions, the label affixed to packages containing highly inflammable insecticides must clearly indicate their flammability. It should warn users to keep the insecticides away from heat sources, open flames, and similar hazards.

Table 9. Classification of insecticides and labeling.

Classification of the Insecticides	Medium Lethal Dose by the Oral Route Acute Toxicity LD 50 mg/kg Body Weight of Test Animals	Medium Lethal Dose by the Dermal Route Dermal Toxicity LD 50 mg/kg Body Weight of Test Animals	Color of Identification Band on the Label
Column-1	Column-2	Column-3	Column-4
1. Extremely toxic	1–50	1–200	Bright red
2. Highly toxic	51–500	201–2000	Bright yellow
3. Moderately toxic	501–5000	2001–20,000	Bright blue
4. Slightly toxic	More than 5000	More than 20,000	Bright green

Source: Insecticide Act of 1968 [23,24].

4.2. Pesticide Residues

The Government of India, through the Ministry of Agriculture & Farmers Welfare, is implementing a central sector scheme known as the “Monitoring of Pesticide Residues at National Level (MPRNL)”. This scheme involves the systematic collection and analysis of samples of vegetables, fruits, and other grains from various sources such as retail outlets, markets, and farm gates. These samples are then analyzed by laboratories accredited by the National Accreditation Board (NAB) for pesticide residues.

During the fiscal year 2017–2018, a total of 27 NAB-accredited laboratories across India participated in the collection and analysis of samples. These laboratories collected and analyzed about 23,660 samples from a diverse range of products, including vegetables, fruits, spices, curry leaves, red chili powder, rice, wheat, pulses, milk, fish/marine products, tea, meat, eggs, and water. The samples were sourced from retail outlets, APMC markets, Mother Dairy outlets, organic outlets, and farm gates [35]. Among these samples, pesticide residues were detected in 19.1% of the total samples. Notably, only 2.2% of the samples contained residues exceeding the maximum residue limit (MRL) set by the FSSAI (Figure 14).

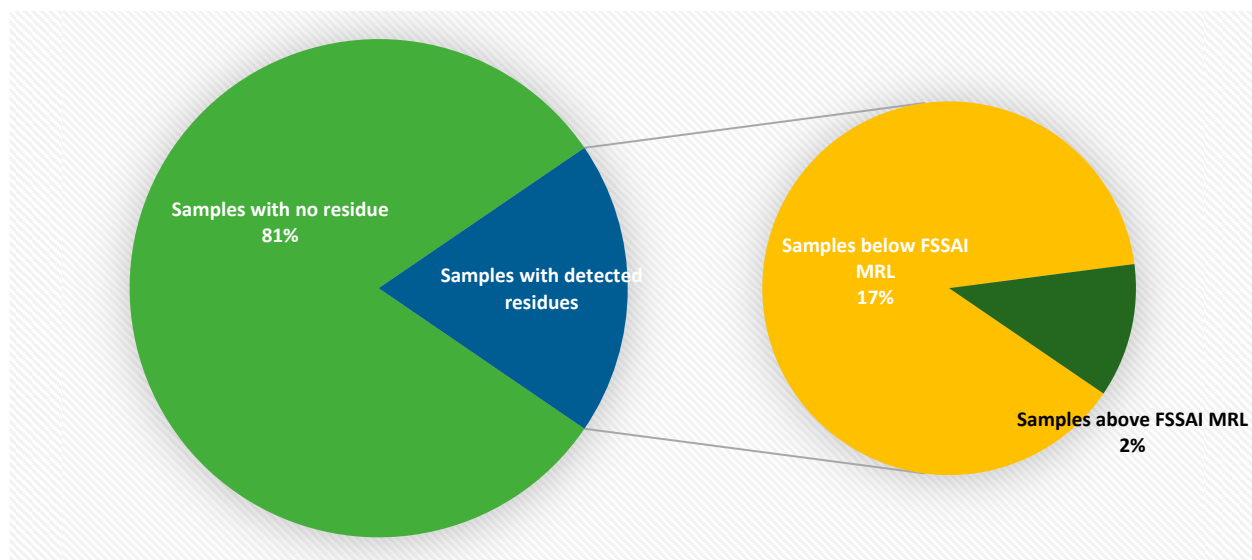


Figure 14. Food samples for pesticide residues. Source: Monitoring of Pesticide Residues at National Level (MPRNL), 2017–2018 report. https://fssai.gov.in/upload/advisories/2019/10/5da705b31ca78Letter_Report_Pesticides_MRL_16_10_2019.pdf (accessed on 1 April 2024).

In the 2017–2018 survey conducted under MPRNL, a total of 12,821 vegetable samples were collected, and a comprehensive analysis was performed on various agricultural products for pesticide residues.

4.2.1. Vegetables

A total of 12,821 vegetable samples, including brinjal, okra, tomato, cabbage, cauliflower, green chili, capsicum, cucumber, green pea, bitter gourd, and coriander leaves, were collected from local retail outlets, markets, farmers' fields (farm gates), and organic outlets. The analysis revealed pesticide residues in 18.7% of vegetable samples, with 1.9% of samples exceeding the MRL prescribed by the FSSAI. Notably, 13.3% of samples contained residues of non-approved pesticides.

4.2.2. Fruits

For fruits, 2274 samples, including apple, banana, pear, grape, orange, pomegranate, guava, sapota, and mango, were analyzed. Findings revealed that although residues were detected in 21.7% of the samples, only 1.1% exceeded FSSAI MRL. No pesticide residues were found in 78.3% of fruit samples, while 12.8% contained non-approved pesticides.

4.2.3. Spices

Among spices, 1242 samples from cardamom, black pepper, cumin, fennel seed, fenugreek seed, dry ginger, coriander seed, and red chili powder were analyzed. Results revealed pesticide residues in 54.8% samples, with 12.1% exceeding the FSSAI MRL. Moreover, in the case of 616 curry leaf samples analyzed, all detected pesticides were non-approved.

4.2.4. Staple Crops

A total of 1960 samples were analyzed. The analysis of rice samples revealed residues in 21.7% of samples, with 7.2% exceeding the FSSAI MRL. In wheat, pesticide residues were found in 9.5% of samples, with 1% exceeding the MRL, and 11.8% of pulse samples had pesticide residues, with just 1.2% surpassing FSSAI MRL.

Moreover, monitoring efforts were extended to tea (180 samples), packaged milk (453 samples), meat/eggs (374 samples), fish/marine products (902 samples), and water (2031 samples), with no samples detected above the MRL. Overall, during the 2019–2020 period, a total of 30,664 samples of fruits, vegetables, and other crops were collected and analyzed for pesticide residues, with banned pesticide residues detected in just 0.25% samples (MPRNL annual report, 2020).

4.3. Ban of Pesticides

The Registration Committee (RC), established by the Indian government, has registered a total of 330 technical and 867 formulated pesticides for use within the country. Among these registrations, 11 technical and 110 formulated pesticides are currently under patent protection. Over the past five years, the RC has registered 44 new pesticides, including 4 under patent [33–35].

Review and banning process: Registered pesticides undergo periodic reviews to assess their safety and efficacy. Upon receiving new studies, reports, references, or information, the government appoints expert committees to conduct these reviews. Based on the recommendations of these committees and after consultation with the RC, the Ministry of Agriculture has taken action to ban or phase out 46 pesticides and four pesticide formulations from import, manufacture, or sale in the country. Additionally, eight pesticide registrations have been withdrawn, and nine pesticides have been placed under restricted use.

Decision-Making Criteria

When making decisions to ban pesticides, stakeholders' objections and suggestions on draft notifications are carefully considered. Factors include technical and scientific requirements, the availability of alternatives, farmers' interests, safety concerns, toxicity, efficacy, and the ban status in other countries. The following pesticides have been banned from 2015 to 2022: alachlor, benomyl, carbaryl, diazinon, dichlorvos, fenarimol, fenthion, linuron,

methoxy ethyl mercury chloride (MEMC), methyl parathion, phorate, phosphamidon, sodium, thiometon, triazophos, tridemorph, and trichlorfon.

4.4. Bio-Pesticides

The Government of India is actively promoting the use of bio-pesticides, which are generally considered safer alternatives to chemical pesticides. A comprehensive list of approved pesticides for various crops is made publicly available on the official website of the Directorate of Plant Protection Quarantine & Storage [35–39].

Moreover, the Central Government is actively encouraging research on the development of bio-pesticides. Guidelines have been established for the registration of bio-pesticide consortia, and fifteen different types of bio-pesticides developed by the ICAR/industry have received approval for pest management across various crops.

Furthermore, the government is implementing the “Sub-Mission on Plant Protection and Plant Quarantine” through 35 Central Integrated Pest Management Centers located across 28 states and two union territories. These centers aim to popularize IPM practices through training and demonstrations. The focus is on promoting biological control approaches in crop protection technology, with an emphasis on minimizing the use of chemical pesticides and ensuring safety. Additionally, cultural, physical, and mechanical pest management methods, as well as the use of bio-pesticides and bio-control agents, are being promoted.

5. Policy Analysis

To enhance crop yields and profitability for farmers, there is an urgent need to optimize the use of approved pesticides while promoting the adoption of bio-pesticides. Pesticide usage varies significantly across different crops and regions. Despite India’s lower use of pesticides per unit area compared to global and developed-country standards, pesticide residues in crop products remain a concern [40]. Presently, a substantial proportion (60–70%) of farmers resort to pesticide usage in pursuit of higher yields under the banner of “modern agriculture”, often at the expense of environmental and health considerations [41]. The combined and interactive impacts of these diverse pesticide products are challenging to gauge accurately for producers, farmers, and policymakers [42]. Consequently, the government faces challenges in keeping pace with the rapid registration of hundreds of pesticides over the years, making it difficult to promptly evaluate, ban, or restrict hazardous chemicals [43].

Pesticide residues sometimes present in crop products at the time of harvest and consumption. The levels of such residues vary across different crops, types of pesticides, and geographical locations [44]. The analysis of residue data from various food items in India, including fruits, vegetables, cereals, pulses, grains, wheat flour, oils, eggs, meat, fish, poultry, bovine milk, butter, and cheese, reveals their presence in at least some instances [45–51].

In India, the production and use of pesticides are governed by a limited number of laws. These laws primarily establish the institutional mechanisms for regulating pesticide activities, encompassing registration procedures, licensing, and quality standards [52–54]. Additionally, they set standards such as maximum residue limits (MRLs) and average daily intake levels to ensure the introduction of chemicals into agricultural practices without compromising environmental integrity or consumer health [55].

5.1. PFA Regulations on Maximum Residue Levels (MRLs)

Historically, the regulation of MRLs for pesticides and agro-chemicals in food products was managed by the Ministry of Health and Family Welfare under the Prevention of Food Adulteration Act (PFA) of 1955. However, with the introduction of the Food Safety and Standards Act (FSSA), 2006, there has been a shift toward the Food Safety and Standards Regulations of 2010. This new legislation empowers the FSSAI to establish limits for a variety of substances. These include food additives, crop contaminants, pes-

ticide residues, residues of veterinary drugs, heavy metals, processing aids, mycotoxins, antibiotics, pharmacological active substances, and the irradiation of food.

The existing MRLs for pesticides and agro-chemicals, initially specified under the PFA, have been integrated into the Food Safety and Standards Regulations of 2010, particularly in Chapter 8, pages 531–548. These MRLs are categorized by chemical product and are assigned to specific food items or commodities. In some cases, tolerance limits are defined for broader categories of food. For example, for the pesticide carbaryl, the MRL for “other vegetables” is set at 5.0 parts per million.

5.2. Regulations on Use of Pesticides

The Ministry of Agriculture regulates the manufacturing, sale, import, export, and use of pesticides in India through the Insecticides Act of 1968 and the Insecticides Rules, 1971. All insecticides, including fungicides and herbicides, listed in the Schedule must be registered by the Central Insecticides Board & Registration Committee (CIB&RC). The list of registered insecticides is available on the CIB&RC website. Registered products must have clear labels indicating their composition, active ingredient(s), target pest(s), recommended dosage, and cautionary safety information. However, these cautionary notes are often printed in small fonts, making them difficult to read.

The Central Insecticides Laboratory (CIL) and Insecticide Inspectors ensure that insecticides sold in the market meet quality standards. The CIL analyzes samples for pesticide residues, efficacy, and toxicity and monitors compliance with registration conditions. Under this legislation, the Central Government handles pesticide registration, while state governments issue marketing licenses. The enforcement of the law is primarily carried out by state agriculture departments.

Officials collaborate with the CIB&RC to analyze pesticide usage, including post-harvest maximum residue limits (MRLs), and enforce public safety regulations. The CIB&RC periodically reviews pesticide usage and may recommend bans on registration if MRLs exceed prescribed limits. A list of banned pesticides is published on the CIB&RC website. The Ministry of Agriculture regularly reviews pesticide safety, and applications may be withdrawn or modified accordingly. In 2006, several pesticide applications were removed from the approved list.

Both central and state governments have the authority to prohibit the sale, distribution, or use of insecticides in specific locations and for specific periods via notification in the official gazette. Additionally, the legislation mandates the reporting of insecticide-related poisonings, which can lead to sales prohibitions if deemed necessary for public safety. The Act imposes penalties for selling misbranded insecticides or contravening licensing requirements. While registration and licensing follow established procedures, the process for banning or prohibiting pesticides is different. India employs lengthy review processes involving specifically convened committees, unlike countries with automatic periodic reviews or bans based on international prohibitions.

6. Conclusions and Future Prospects

This study underscores that raising awareness among farmers about cost-effective and environmentally friendly pesticide use is crucial to optimize crop yields and enhance profitability [56]. Efforts must focus on not only the judicious use of approved pesticides but also encouraging the adoption of bio-pesticides to achieve a balance between high agricultural productivity and minimal environmental impact [57]. Immediate improvements to the domestic pesticide industry are essential, both to meet the critical needs of farmers and to maintain India's position as a significant exporter of pesticides [58]. Figure 15 presents a theory of change, illustrating how the proper implementation of the Pesticides Regulation Act, coupled with farmer education and awareness, can positively impact farmers' incomes and the bio-environment.

Strategic pesticide use ensures that pesticides are used judiciously and only when necessary, thereby reducing the overall quantity applied [57]. Moreover, the development

and use of bio-pesticides is vital for environmental and human safety [59]. Training farmers in safe pesticide practices and alternatives such as integrated pest management (IPM) practices is essential [60–62].

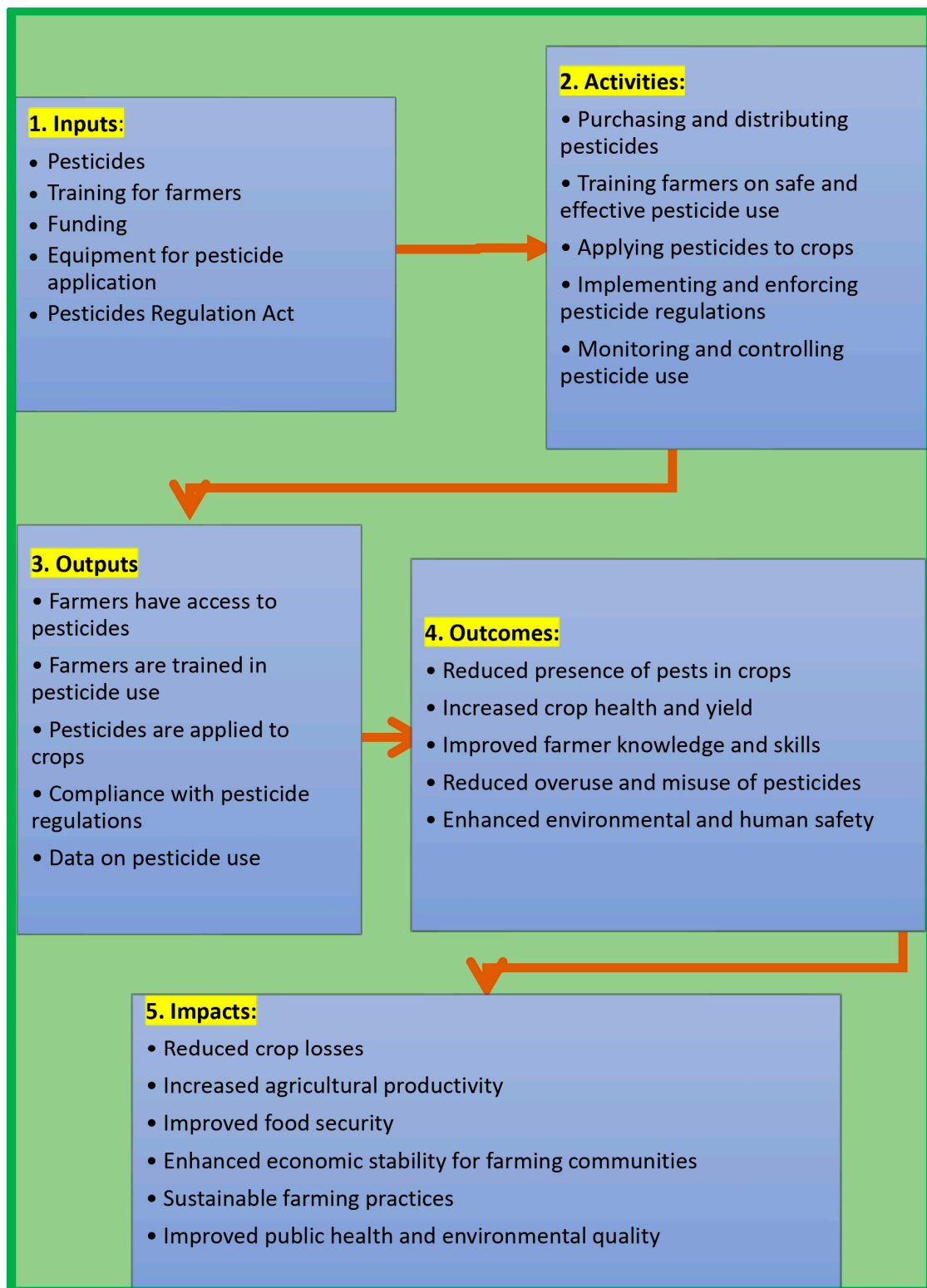


Figure 15. Theory of change with the proper development of the Pesticides Regulation Act.

Moreover, improving testing and regulatory reforms, such as upgrading laboratory capabilities, expanding testing parameters, and conducting regular reviews, are required to

promptly ban hazardous pesticides and ensure the integrity of the pesticide supply chain, eliminating counterfeit products based on the latest scientific evidence [63,64]. Rigorous quality assurance at the point of sale and mechanisms to safeguard farmers are essential to eliminate counterfeit products from the supply chain [65]. Enhancing monitoring and compliance mechanisms will ensure adherence to established pesticide regulations and standards [3].

Accurate data is vital for understanding and managing the risks associated with pesticide use. Therefore, addressing the data gaps on pesticide production and usage is essential. Reconciling the production, consumption, and trade data of chemical pesticides from various sources remains challenging and requires focused attention [66]. By integrating these strategies, a more sustainable agricultural framework can be established, balancing productivity with safety and environmental conservation [67–71].

Author Contributions: Conceptualization, A.A.R., M.R. and V.M.; methodology, A.A.R., M.R. and V.M.; validation, V.M.; formal analysis, V.M.; investigation, M.R.; data curation, A.A.R. and M.R.; writing—original draft preparation, M.R.; writing—review and editing, V.M.; visualization, V.M.; supervision, A.A.R. and V.M.; Project administration, A.A.R. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The original contributions presented in the study are included in the article, further inquiries can be directed to the corresponding authors.

Conflicts of Interest: The authors declare no conflict of interest.

Appendix A

Table A1. Major insect pests, damage, yield loss, and monetary potential of crops.

S. No.	Crop	Major Pests	Damage (%)	Yield Loss (%)	Yield (kg/ha)	Monetary Potential (Rs/ha)	Loss Avoidance Potential (Minimum) (Rs/ha)	Loss Avoidance Potential (Maximum) (Rs/ha)
1.	Paddy	Yellow stem borer, <i>Scirpophaga incertulas</i>	10–20	25–30	2404	52,479	13,120	15,744
		Brown plant hopper, <i>Nilaparvata lugens</i>	40–50	10–70	2404	52,479	5248	36,735
		Gall midge, <i>Orseolia oryzae</i>	15	70–85	2404	52,479	36,735	44,607
		Leaf folder, <i>Cnaphalocrocis medinalis</i>	1–30	40–57	2404	52,479	20,992	29,913
2.	Cotton	Leafhopper, <i>Amrasca devastans</i>	40–50	30–35	445	29,459	8838	10,311
		Whitefly, <i>Bemisia tabaci</i>	40	15–30	445	29,459	4419	8838
		Tobacco caterpillar, <i>Spodoptera litura</i>	30–40	30–40	445	29,459	8838	11,784
		Pink bollworm, <i>Pectinophora gossypiella</i>	20–80	20–95	445	29,459	5892	27,986
		Spotted and spiny bollworm, <i>Earias vittella</i> , <i>E. insulana</i>	30–40	30–40	445	29,459	8838	11,784
		American bollworm, <i>Helicoverpa armigera</i>	20–30	20–80	445	29,459	5892	23,567

Table A1. Cont.

S. No.	Crop	Major Pests	Damage (%)	Yield Loss (%)	Yield (kg/ha)	Monetary Potential (Rs/ha)	Loss Avoidance Potential (Minimum) (Rs/ha)	Loss Avoidance Potential (Maximum) (Rs/ha)
3.	Sugarcane	Early shoot borer, <i>Chilo infuscatellus</i>	Medium	20–25	84,000	285,600	57,120	71,400
		Pink stem borer, <i>Sesamia inferens</i>	29.40	55–60	84,000	285,600	157,080	171,360
		Top shoot borer, <i>Scirpophaga excerptalis</i>	Medium	21–37	84,000	285,600	59,976	105,672
		Pyrilla, <i>Pyrilla purpusilla</i>	Medium	30–35	84,000	285,600	85,680	99,960
		Woolly aphid, <i>Ceratovacuna lanigera</i>	100	50–55	84,000	285,600	142,800	157,080
		Internode borer, <i>Chilo sacchariphagus indicus</i>	80	80–85	84,000	285,600	228,480	242,760
4.	Chili	Tobacco cut worm, <i>Spodoptera litura</i>	2–8	30–40	12,000	819,960	245,988	327,984
		Gram pod borer, <i>Helicoverpa armigera</i>	High	77–75	12,000	819,960	631,369	614,970
		Chili black thrips, <i>Thrips parvispinus</i>	high	50–80	12,000	819,960	409,980	655,968
		Whitefly, <i>Bemisia tabaci</i>	High	30–40	12,000	819,960	245,988	327,984
		Yellow mite, <i>Polyphagotarsonemus latus</i>	Medium to high	30–50	12,000	819,960	245,988	409,980

Note: MSP 2023–2024: @Rs. 2183/q (rice), cotton @Rs. 6620/q, sugarcane @Rs. 340/q, and chili @Rs. 6833/q (market price). Calculated from [72–75].

Table A2. Major diseases, damage, yield loss, and monetary potential of crops.

S. No.	Crop	Major Pests	Damage (%)	Yield Loss (%)	Yield (kg/ha)	Monetary Potential (Rs/ha)	Loss Avoidance Potential (Minimum)	Loss Avoidance Potential (Maximum)
1.	Paddy	Blast, <i>Pyricularia oryzae</i> (<i>Magnaporthe oryzae</i>)	Low to high	70–80	2404	52,479	36,735	41,983
		Bacterial leaf blight, <i>Xanthomonas oryzae</i> pv. <i>oryzae</i>	Low to high	50–80	2404	52,479	26,240	41,983
		Brown spot, <i>Bipolaris oryzae</i>	Low to high	26–52	2404	52,479	13,645	27,289
		Sheath blight, <i>Rhizoctonia solani</i>	Low to high	45–55	2404	52,479	23,616	28,863
		Sheath rot, <i>Sarocladium oryzae</i>	Low to high	5–80	2404	52,479	2624	41,983
2.	Cotton	Leaf curl, cotton leaf curl virus	100	85–95	445	29,459	25,040	27,986
		Angular leaf spot/BLB, <i>Xanthomonas axonopodis</i> pv. <i>Malvacearum</i>	26–55	5–35	445	29,459	1473	10,311
		Alternaria blight, <i>Alternaria gossypina</i> , <i>A. alternata</i>	24–40	26.60	445	29,459	5892	8838
		Myrothecium leaf spot, <i>Myrothecium roridum</i>	34	25–60	445	29,459	7365	17,675

Table A2. Cont.

S. No.	Crop	Major Pests	Damage (%)	Yield Loss (%)	Yield (kg/ha)	Monetary Potential Yield (Rs/ha)	Loss Avoidance Potential (Minimum)	Loss Avoidance Potential (Maximum)
3.	Sugarcane	Red rot, <i>Colletotrichum falcatum</i>	High in sub-tropical areas	50–100	84,000	285,600	142,800	2,85,600
		Smut, <i>Sporisorium scitamineum</i>	High in sub-tropical areas	25–50	84,000	285,600	71,400	142,800
		Wilt, <i>Fusarium sacchari</i>	High	15–20	84,000	285,600	42,840	57,120
		Grassy shoot disease, SCGS Phytoplasma	High	5–70	84,000	285,600	14,280	199,920
4.	Chili	Powdery mildew, <i>Leveillula taurica</i>	10–20	14–30	12,000	819,960	1,14,794	245,988
		Die back and fruit rot, <i>Colletotrichum capsici</i>	25–47	10–50	12,000	819,960	81,996	409,980
		Leaf curl, Begomovirus	High	50–100	12,000	819,960	409,980	819,960
		Alternaria leaf spot, <i>Alternaria solani</i>	High	50–100	12,000	819,960	409,980	819,960

Notes: MSP 2023–2024 @Rs. 2183/q (rice), cotton @Rs. 6620/q, sugarcane @Rs. 340/q, and chili @Rs. 6833/q. Calculated from [72–75].

Table A3. Potential, actual and monetary yield losses (%) due to weeds in major field crops.

S. No.	Crop	Yield Loss Potential (%)	Yield (kg/ha)	Monetary Potential Yield (Rs/ha)	Loss Avoidance Potential (Minimum)	Loss Avoidance Potential (Maximum)
1.	Rice	10–100	2404	2404	52,479	5248
3.	Sugarcane	25–50	84,000	84,000	285,600	71,400
4.	Cotton	40–60	445	445	29,459	11,784
5.	Chili	60–80	12,000	12,000	819,960	491,976

Note: Calculated from [72–75].

References

- Reddy, A.A.; Bhattacharya, A.; Reddy, S.V.; Ricart, S. Farmers' distress index: An approach for an action plan to reduce vulnerability in the drylands of India. *Land* **2021**, *10*, 1236. [CrossRef]
- Oerke, E.C. Crop losses to pests. *J. Agric. Sci.* **2006**, *144*, 31–43. [CrossRef]
- Dhaliwal, G.S.; Jindal, V.; Mohindru, B. Crop losses due to insect pests: Global and Indian scenario. *Indian J. Entomol.* **2015**, *77*, 165–168. [CrossRef]
- Karthikeyan, L.; Chawla, I.; Mishra, A.K. A review of remote sensing applications in agriculture for food security: Crop growth and yield, irrigation, and crop losses. *J. Hydrol.* **2020**, *586*, 124905. [CrossRef]
- Sawicka, B.; Egbuna, C. Pests of Agricultural Crops and Control Measures. In *Natural Remedies for Pest, Disease and Weed Control*; Academic Press: Cambridge, MA, USA, 2020; pp. 1–16.
- Kumar, V.; Khan, M.R.; Walia, R.K. Crop loss estimations due to plant-parasitic nematodes in major crops in India. *Natl. Acad. Sci. Lett.* **2020**, *43*, 409–412. [CrossRef]
- Nayak, P.; Solanki, H. Pesticides and Indian agriculture—are view. *Int. J. Res. Granthaalayah* **2021**, *9*, 250–263. [CrossRef]
- Kumar, S.; Sharma, A.K.; Rawat, S.S.; Jain, D.K.; Ghosh, S. Use of pesticides in agriculture and livestock animals and its impact on environment of India. *Asian J. Environ. Sci.* **2013**, *8*, 51–57.
- Peshin, R.; Hansra, B.S.; Nanda, R.; Singh, K.; Sharma, R.; Garg, L.; Bajiya, M.R.; Showkat, A.; Kumar, R.; Yangsdon, S. Pesticides hazardous hotspots: Empirical evidences from North India. *Environ. Manag.* **2020**, *66*, 899–915. [CrossRef]
- Alavanja, M.C. Introduction: Pesticides use and exposure, extensive worldwide. *Rev. Environ. Health* **2009**, *24*, 303–310. [CrossRef]
- Available online: <https://www.fao.org/3/cc0918en/cc0918en.pdf> (accessed on 1 March 2024).
- Abhilash, P.C.; Singh, N. Pesticide use and application: An Indian scenario. *J. Hazard. Mater.* **2009**, *165*, 1–12. [CrossRef]
- Devi, P.I.; Thomas, J.; Raju, R.K. Pesticide consumption in India: A spatiotemporal analysis. *Agric. Econ. Res. Rev.* **2017**, *30*, 163–172. [CrossRef]
- Koli, P.; Bhardwaj, N.R. Status and use of pesticides in forage crops in India. *J. Pestic. Sci.* **2018**, *43*, 225–232. [CrossRef]
- Shetty, P.K. Socio-ecological implications of pesticide use in India. *Econ. Political Wkly.* **2004**, *39*, 5261–5267.

16. FSSAI 2019. Available online: https://fssai.gov.in/upload/advisories/2019/10/5da705b31ca78Letter_Report_Pesticides_MRL_16_10_2019.pdf (accessed on 1 January 2024).
17. Agriculture Census Division. *All India Report on Input Survey 2011–2012*; Department of Agriculture, Cooperation & Farmers Welfare: New Delhi, India, 2016.
18. Agriculture Census Division. *All India Report on Input Survey 2016–2017*; Department of Agriculture, Cooperation & Farmers Welfare: New Delhi, India, 2021.
19. Available online: <https://www.fao.org/faostat/en/#data/RP> (accessed on 1 January 2024).
20. Available online: https://ppqs.gov.in/sites/default/files/graphs_0.pdf (accessed on 1 January 2024).
21. Available online: <https://sansad.in/getFile/loksabhaquestions/annex/1710/AU1069.pdf?source=pqals> (accessed on 1 March 2024).
22. States/UTs Zonal Conferences on Inputs (Plant Protection) for Rabi & Kharif Seasons. Available online: <https://ppqs.gov.in> (accessed on 1 March 2024).
23. Available online: <https://www.indiacode.nic.in/bitstream/123456789/1551/1/A1968-46.pdf> (accessed on 1 March 2024).
24. Available online: https://upload.indiacode.nic.in/showfile?actid=AC_CEN_23_31_00001_196846_1517807318487&type=rule&filename=Insecticides%20Rule,%201971.pdf (accessed on 1 June 2024).
25. Available online: <https://ppqs.gov.in/divisions/central-insecticides-board-registration-committee> (accessed on 1 July 2024).
26. Available online: https://ppqs.gov.in/sites/default/files/pesticide_formulations_registered_for_use_in_the_country.pdf (accessed on 1 August 2024).
27. Pandey, J.K. Use of Hazardous Chemical Pesticides in India: A Review. *Proc. Natl. Acad. Sci. India Sect. B Biol. Sci.* **2023**, *93*, 523–537. [CrossRef]
28. Dubey, J.K.; Sharma, A. Pesticide Residues and International Regulations. In *Sustainable Management of Potato Pests and Diseases*; Springer: Singapore, 2022; pp. 353–367.
29. Zaller, J.G.; Zaller, J.G. Where Are the Solutions to the Pesticide Problem? In *Daily Poison: Pesticides-an Underestimated Danger*; Springer: Cham, Switzerland, 2020; pp. 223–295.
30. Kuchheuser, P.; Birringer, M. Pesticide residues in food in the European Union: Analysis of notifications in the European Rapid Alert System for Food and Feed from 2002 to 2020. *Food Control* **2022**, *133*, 108575. [CrossRef]
31. FICCI. *Ushering in the 2nd Green Revolution: Role of Crop Protection Chemicals*; Federation of Indian Chambers of Commerce and Industry: New Delhi, India, 2015.
32. Patel, P.; Naik, A. Ethical and Legal Issues in Microbial Products in India. In *Agricultural Microbiology Based Entrepreneurship: Making Money from Microbes*; Springer Nature: Singapore, 2022; pp. 81–100.
33. Birthal, P.S. *Economic Potential of Biological Substitutes for Agrochemicals*; Policy Paper 18; National Centre for Agricultural Economics and Policy Research (ICAR): New Delhi, India, 2003.
34. Narayanamoorthy, A. Profitability in crops cultivation in India: Some evidence from cost of cultivation survey data. *Indian J. Agric. Econ.* **2013**, *68*, 104–121.
35. ICAR 2024. Available online: <https://aicrp.icar.gov.in/pesticide/monitoring-of-pesticide-residues/> (accessed on 1 August 2024).
36. Available online: https://agcensus.gov.in/AgriCensus/Agri_2122.jsp# (accessed on 1 August 2024).
37. Available online: <https://ppqs.gov.in/statistical-database> (accessed on 20 August 2024).
38. Available online: <https://ppqs.gov.in/divisions/cib-rc/registered-products> (accessed on 20 August 2024).
39. Reddy, A.A.; Melts, I.; Mohan, G.; Rani, C.R.; Pawar, V.; Singh, V.; Choubey, M.; Vashishtha, T.; Suresh, A.; Bhattarai, M. Economic impact of organic agriculture: Evidence from a Pan-India survey. *Sustainability* **2022**, *14*, 15057. [CrossRef]
40. Kumar, R. Pesticide use in modern agriculture: Impacts on environment and human health. *J. Agric. Sci.* **2014**, *10*, 45–57.
41. Patel, A.; Mehta, S. Challenges in pesticide regulation and monitoring. *Environ. Policy Rev.* **2018**, *12*, 120–135.
42. Sharma, V.; Singh, A.; Gupta, R. The complexity of pesticide interactions and regulatory challenges. *Pestic. Sci. Manag.* **2016**, *8*, 67–78.
43. Singh, M. Pesticide residues in Indian agriculture: An overview. *Agric. Res. J.* **2012**, *15*, 89–98.
44. Gupta, R.; Sharma, V.; Singh, A. Monitoring pesticide residues in food products: A case study from India. *Int. J. Food Saf.* **2017**, *31*, 78–89.
45. Kumar, A.; Singh, B. Pesticide residue analysis in food items of India. *Food Saf. J.* **2013**, *25*, 112–121.
46. Mehta, S. The regulatory landscape of pesticide use in India: Challenges and prospects. *J. Agric. Policy* **2018**, *21*, 145–160.
47. Patel, R. Regulation of pesticide production and use in India. *J. Environ. Law* **2016**, *14*, 54–67.
48. Rao, S.; Reddy, P. Standards for pesticide residues in agricultural practices. *Agric. Regul. Rev.* **2014**, *18*, 102–110.
49. Sankaran, S.; Hegde, N. Geographic and crop-specific variations in pesticide residues. *Indian J. Agric. Sci.* **2010**, *20*, 189–198.
50. Sharma, V.; Kumar, R.; Patel, A. Residue levels of pesticides in Indian agricultural produce: Current scenario and future perspectives. *Pestic. Res. J.* **2015**, *27*, 92–105.
51. Singh, R.; Mehta, S.; Gupta, K. Pesticide residue trends in Indian food: A comprehensive review. *J. Food Qual.* **2020**, *40*, 211–224.
52. Verma, D.; Sharma, S.; Jain, R. Assessment of pesticide residues in food commodities from different regions of India. *J. Environ. Monit.* **2019**, *22*, 157–168.
53. Vikas, P.; Nair, A.; Dubey, S. Recent advances in pesticide residue detection and management in India. *Curr. Opin. Environ. Sci. Health* **2021**, *25*, 135–144.

54. Yadav, S.; Gupta, R.; Mehta, P. Evaluating pesticide residues in Indian agriculture: Policy implications and future strategies. *Agric. Policy Res. J.* **2018**, *19*, 121–133.
55. Zafar, A.; Khan, N.; Sinha, P. Regulatory measures for pesticide residues in India: An analysis of current practices and recommendations. *Environ. Policy Gov.* **2017**, *15*, 76–85.
56. Dasgupta, S.; Meisner, C.; Huq, M. A pinch or a pint? Evidence of pesticide overuse in Bangladesh. *J. Agric. Econ.* **2007**, *58*, 91–114. [[CrossRef](#)]
57. Fernandez-Cornejo, J. Environmental and economic consequences of technology adoption: IPM in viticulture. *Agric. Econ.* **1998**, *18*, 145–155. [[CrossRef](#)]
58. Kumar, S. Pesticide industry in India: Progress and potential. *J. Plant Prot. Sci.* **2011**, *3*, 1–13.
59. Peshin, R.; Bandral, R.S.; Zhang, W.; Wilson, L.; Dhawan, A.K. Integrated Pest Management: A global Overview of History, Programs and Adoption. In *Integrated Pest Management: Innovation-Development Process*; Peshin, R., Dhawan, A.K., Eds.; Springer: Dordrecht, The Netherlands, 2009; Volume 1, pp. 1–49.
60. Popp, J.; Pető, K.; Nagy, J. Pesticide productivity and food security. *A Review. Agron. Sustain. Dev.* **2013**, *33*, 243–255. [[CrossRef](#)]
61. Pretty, J. Agricultural sustainability: Concepts, principles and evidence. *Philos. Trans. R. Soc. B Biol. Sci.* **2008**, *363*, 447–465. [[CrossRef](#)]
62. Reganold, J.P.; Glover, J.D.; Andrews, P.K.; Hinman, H.R. Sustainability of three apple production systems. *Nature* **2011**, *410*, 926–930. [[CrossRef](#)]
63. Aktar, M.W.; Sengupta, D.; Chowdhury, A. Impact of pesticides use in agriculture: Their benefits and hazards. *Interdiscip. Toxicol.* **2009**, *2*, 1–12. [[CrossRef](#)]
64. Bhardwaj, T.; Sharma, J.P.; Bhardwaj, A.K. Pesticide residue management in Indian agriculture: A review. *Int. J. Curr. Microbiol. Appl. Sci.* **2018**, *7*, 2547–2563.
65. De, A.; Debnath, A.; Chowdhury, R. Pesticide applications—Contemporary trends and emerging issues. *Int. J. Environ. Sci. Technol.* **2014**, *11*, 2245–2252.
66. Jeyaratnam, J. Acute pesticide poisoning: A major global health problem. *World Health Stat. Q.* **1990**, *43*, 139–144. [[PubMed](#)]
67. Kumari, B.; Madan, V.K. Contamination of farmgate vegetables with pesticide residues in Haryana, India. *Environ. Monit. Assess.* **2020**, *192*, 1–13.
68. Nathan, H. Current status of pesticide regulations and their implementation in India. *J. Regul. Sci.* **2018**, *6*, 1–10.
69. Pimentel, D.; Burgess, M. Environmental and economic costs of the application of pesticides primarily in the United States. *Integr. Pest Manag.* **2014**, *3*, 47–71.
70. Shetty, P.K.; Hiremath, M.B.; Murugan, M.; Rao, P. Farmers’ health issues due to pesticide exposure: Medical and economic analysis. *J. Agromedicine* **2017**, *22*, 118–130.
71. Reddy, A.A. The soil health card Scheme in India: Lessons learned and challenges for replication in other developing countries. *J. Nat. Resour. Policy Res.* **2019**, *9*, 124–156. [[CrossRef](#)]
72. Murali Baskaran, R.K.; Jain, S.K.; Sivalingam, P.N.; Mooventhana, P.; Sridhar, J.; Sharma, K.C.; Kaushal, P.; Kumar, J. *National Status of Biotic Stresses of Crops*; ICAR—National Institute of Biotic Stress Management: Raipur, India, 2019; 174p.
73. Gharde, Y.; Singh, P.K. *Yield and Economic Losses Due to Weeds in India (2018)*; ICAR—Directorate of Weed Research: Jabalpur, India, 2018; 22p.
74. Rao, A.N.; Wani, S.P.; Ladha, J.K. Weed management research in India—An analysis of the past and outlook for future. In *Souvenir (1989–2014)*; DWR publication No. 18; Directorate of Weed Research: Jabalpur, India, 2014; pp. 1–26.
75. Sharma, S.; Sharma, A.K.; Jaipaul; Bhatta, M.P. Genotypic differences in growth, yield and quality attributes of capsicum under black polyethylene mulch. *Indian J. Agric. Sci.* **2011**, *81*, 413–416.

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.